Demand for Rare Success: Salient Realizations and Latent Quality in the Stallion Market

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Abstract

In many markets, buyers choose among differentiated products whose value depends on latent, product-specific probabilities of rare post-purchase success. We study such demand in the Japanese thoroughbred stallion market, where breeders choose stallions before learning whether the resulting offspring will become elite racehorses. Using a 2004–2024 stallion-year panel and an IV pedigree-nested logit demand system, we distinguish latent quality—an empirical-Bayes, exposure-adjusted measure of a stallion's ability to produce elite offspring—from recent graded-stakes victories by the stallion's progeny, salient public realizations of rare success. These victories shift demand substantially, yet add little to forecasts of future elite output once latent quality is accounted for. For stallions at representative baseline shares, a recent G1 victory raises the predicted share of matings by roughly 80 percent, a demand effect comparable to moving the stallion from the 25th to the 75th percentile of latent quality. An output-alignment exercise shows that, holding other breeders' choices fixed, a representative breeder would obtain a higher predicted probability of elite offspring by relying more on latent quality and less on salient success signals. The same signals are reflected in posted stud fees and redirect demand across pedigree-structured substitutes. The findings highlight a demand channel in markets where rare-success probabilities are latent: salient public realizations can carry market value beyond their incremental forecasting content.

Keywords: rare-success demand; salient signals; latent quality; empirical Bayes; differentiated products; stallion market
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1 Introduction

A common feature of many markets is that buyers must choose before learning whether a rare and valuable outcome will materialize. Such environments arise when buyers choose

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products, projects, or talent whose value depends on exceptional outcomes that are revealed only later. The relevant choice is a chance at rare success, and that chance differs across products in ways buyers cannot directly observe. Buyers must therefore infer latent success probabilities from sparse signals. A recent public realization of rare success can then play two roles: it may reveal information about latent quality, but it may also attract demand because it is a salient realization of the outcome buyers hope to obtain.

This paper studies this problem in the Japanese thoroughbred stallion market. Each breeding season, breeders choose among active stallions before learning whether the resulting offspring will become elite racehorses. Stallions differ in *observable attributes*—including pedigree, posted stud fees, own racing records, service age, and accumulated progeny performance—and in the *latent probability* that a mating produces elite offspring. Some public attributes, especially the stallion's own racing credentials, provide initial information about this probability; progeny outcomes provide sparse evidence as cohorts mature. The probability itself is not directly observed, so the setting is a differentiated market rather than a homogeneous lottery. The setting is empirically useful because the rare outcome is public and well defined. In thoroughbred racing, graded-stakes victories are highly visible and commercially meaningful achievements. They are also typically featured in breeding-season materials alongside stud fees, the stallion's own racing record, and pedigree. A graded-stakes victory by a stallion's progeny is therefore both a realized elite outcome and a salient public signal of the rare-success outcome valued in the market.

A central empirical challenge is that the rare-success probability relevant for demand is latent. A useful modeling parallel comes from insurance demand, where consumers choose before knowing whether a claim will arrive and researchers model claim probabilities as latent arrival rates inferred from sparse realizations. For example, Barseghyan et al. (2013) use a Poisson-Gamma structure to measure latent claim rates that enter insurance choices. Our setting has the opposite payoff sign: breeders choose stallions for the chance to obtain a rare upside outcome rather than protection against a rare loss. We use an analogous measurement logic to construct an empirical-Bayes measure of latent quality. Predetermined public characteristics, especially the stallion's own racing record, form an initial assessment, and exposure-adjusted progeny outcomes that accumulate over time update that assessment. This state variable summarizes accumulated information about

rare-success probabilities before the breeding choice. We then ask whether recent public realizations shift demand beyond what is captured by latent quality, and whether the induced demand is aligned with future elite output.

Guided by this measurement logic, our empirical design separates two objects that are often difficult to distinguish when rare-success probabilities are latent. The first is latent quality: an empirical-Bayes measure of a stallion's exposure-adjusted ability to produce elite offspring. This measure summarizes accumulated progeny outcomes while accounting for differences in mating exposure. The second is a salient recent success signal: a recent graded-stakes victory by the stallion's progeny. Such victories are public, easy to interpret, and commercially meaningful. They may convey information about latent quality, but they may also attract demand because they are salient public realizations of the rare outcome breeders hope to obtain.

We bring these objects into a differentiated-products demand framework using a 2004–2024 stallion-year panel from the Japanese thoroughbred breeding market. The posted-fee sample contains 4,467 stallion-year observations for 833 stallions. We estimate an IV pedigree-nested logit demand system that treats stallions as differentiated options within each breeding season, accounts for price and within-nest-share endogeneity, and allows substitution to be locally structured by pedigree.

The IV demand estimates show that breeders respond strongly to both latent quality and salient recent success signals. After controlling for posted fees, own racing records, service age, and latent quality, and allowing for pedigree-based substitution, recent graded-stakes victories by a stallion's progeny remain large demand shifters. For stallions at representative baseline shares, a recent G1 victory raises the predicted share of matings by roughly 80 percent, a demand effect comparable to moving the stallion from the 25th to the 75th percentile of latent quality. Thus, a recent G1 progeny victory is not merely a statistically detectable signal; it is an economically large demand shifter.

We next ask whether the demand response to recent victories reflects incremental forecasting content for future elite output. The empirical-Bayes latent-quality measure robustly predicts future elite output from the subsequent mating cohort, but recent graded-stakes victories add little to those forecasts once latent quality and durable stallion characteristics are accounted for. This contrast motivates an output-alignment exercise: holding other

breeders' choices fixed, a representative breeder would obtain a higher predicted probability of elite offspring by relying more on latent quality and less on salient success signals. The same signals are reflected in posted stud fees and redirect demand across pedigree-structured substitutes. Together, the results show that salient public realizations can carry market value beyond their incremental forecasting content in markets where rare-success probabilities are latent.

This paper contributes to research on decision-making when payoff-relevant event probabilities are small, latent, and learned from sparse realizations. Insurance-demand studies provide the closest downside benchmark, modeling latent claim probabilities and documenting how small-probability risks shape choices (Cohen and Einav, 2007; Sydnor, 2010; Barseghyan et al., 2013). Related field evidence shows that recent realizations of rare adverse events can shift insurance demand (Gallagher, 2014). A second nearby literature studies horse-race betting and parimutuel markets, where odds and outcomes are used to infer risk preferences, probability perceptions, belief heterogeneity, and information aggregation (Thaler and Ziemba, 1988; Jullien and Salanié, 2000; Gandhi and Serrano-Padial, 2015; Chiappori et al., 2019). Information-based models and recent evidence from Japanese interim odds further show that market prices may not fully summarize probability-relevant information when betting dynamics or information arrival matter (Ottaviani and Sørensen, 2009, 2010; Hanyu et al., 2025). We study a different market within the same broad rare-event environment: breeders choose among differentiated stallions for the chance to produce a rare desirable outcome. Our measurement object follows the same rare-event logic, but the economic question differs. We ask whether salient realizations of rare success move demand beyond an empirical-Bayes state variable summarizing the latent probability of producing elite offspring.

The paper also connects to research on salience, consumer learning, and public quality information. Models of salience show how vivid attributes or outcomes can receive disproportionate weight in risky and consumer choice (Bordalo et al., 2012, 2013), and empirical work shows that salience can shape market behavior (Chetty et al., 2009). Consumer-learning models study how buyers update about latent product attributes (Erdem and Keane, 1996; Crawford and Shum, 2005), while work on public quality information and rankings studies how markets respond to disclosed or summarized quality signals (Jin and

Leslie, 2003; Dranove and Jin, 2010; Luca and Smith, 2013). Our contribution is to compare the demand value of a public signal with its subsequent forecasting content: recent graded-stakes victories are realized rare successes whose demand impact can be evaluated against future elite output. This distinction is related to work separating informative and prestige effects of advertising (Ackerberg, 2001). Relatedly, the marketing literature on probabilistic selling studies settings in which sellers intentionally leave the exact product realization or product attributes unresolved at the time of purchase (Fay and Xie, 2008, 2010). Our setting is different: the stallion identity is observed, while the stallion-specific probability of producing future rare success is latent and learned over time from progeny outcomes.

Finally, the paper connects this behavioral and informational question to empirical IO demand estimation. Behavioral industrial organization studies how behaviorally motivated consumer perceptions and choices affect market outcomes (Heidhues and Kőszegi, 2018). Our approach is empirical rather than a full behavioral structural model: we place a salience question inside a differentiated-products demand framework. Recent public realizations of rare success enter demand separately from a latent-quality state, allowing us to ask whether those realizations shift demand beyond accumulated information about rare-success probabilities and whether the shifts are aligned with future output. The demand system follows the empirical IO tradition of estimating substitution and price sensitivity in differentiated markets while addressing price endogeneity (Berry, 1994; Berry et al., 1995; Nevo, 2001). We use pedigree-based nests and differentiation instruments to allow local substitution among related stallions and to account for endogenous posted fees and within-nest shares (Grigolon and Verboven, 2014; Gandhi and Houde, 2019). This structure lets us quantify how salient success signals shift demand, how those gains are diverted across substitutes, and how posted fees respond to the same public realizations.

The remainder of the paper proceeds as follows. Section 2 describes the market setting and the rare-success environment. Section 3 presents the data, market definition, and construction of the empirical-Bayes latent quality measure. Section 4 introduces the nested-logit demand model and identification strategy. Section 5 reports demand estimates and economic magnitudes. Section 6 studies the forecasting content of recent graded-stakes victories and presents the output-alignment exercise. Section 7 examines capitalization

into posted fees and the diversion of signal-induced demand across pedigree-structured substitutes. Section 8 discusses robustness and scope. Section 9 concludes.

2 Market Setting and Information Environment

This section describes why the Japanese thoroughbred breeding market provides a useful field setting for studying demand when rare-success probabilities are latent. Three institutional features are central. First, thoroughbred breeding is registry-based: pedigree, breeding, and offspring records leave detailed administrative traces. These records make it possible to connect stallion choices in a breeding season to subsequent offspring outcomes. Second, breeders choose among high-value, differentiated stallions whose economic value depends on the future racing success of their offspring. Third, racing outcomes are publicly recorded in detail, so the market provides both pre-choice public signals and later realized outcomes. The empirical design uses these features to link public information, posted stud fees, mating choices, and future elite output.

2.1 Breeding-season choices and rare-success payoffs

Each breeding season, breeders choose among active stallions when deciding where to mate their mares. The economic product in the demand analysis is a stallion in a breeding season, or a stallion-year. A breeder's choice determines the stallion used for a mating, but the economically valuable outcome is realized only later, after the resulting offspring is born, matures, and competes in races. The relevant payoff is therefore delayed and uncertain: a mating may eventually produce an elite racehorse, but this rare outcome is not known at the time of choice.

The rare-success outcome in this paper is elite progeny performance in graded-stakes races. Japanese thoroughbred racing classifies elite races into graded categories, with G1 races at the top, followed by G2 and G3 races. A first G1, G2, or G3 victory by a stallion's offspring is a public and commercially meaningful realization of elite performance. These events are rare relative to the number of mating opportunities, but they are central to the value breeders seek when choosing stallions.

This setting contains the main objects needed for the analysis: registered mating choices, posted stud fees, public progeny-performance signals, pedigree-based differenti-

ation, and later rare-success outcomes. The delayed payoff structure also motivates the paper's distinction between latent quality and recent public success signals. Latent quality summarizes accumulated information about a stallion's ability to convert mating opportunities into elite offspring. Recent graded-stakes victories by the stallion's progeny are different: they are newly visible public realizations of the rare outcome itself. Such victories may be informative about the stallion's underlying rare-success probability, but they may also attract demand because they are salient instances of the outcome breeders hope to obtain.

2.2 Timing of public signals, posted fees, and mating choices

The breeding market has a seasonal timing structure that is central to the empirical design. In Japan, the official thoroughbred breeding season runs from February 10 through June 30. A successful mating is followed by an approximately eleven-month gestation period, so the resulting foal is born in the following calendar year. Because thoroughbred age is assigned by birth year rather than by exact birth date within the year, breeders have a strong incentive to arrange matings early in the breeding season: a foal born earlier in the calendar year has more time to grow, train, and prepare before competing in age-restricted races as a two- or three-year-old.

The annual index t in the demand panel refers to the breeding season. The recent-success variables are aligned with the most recent completed racing year before that breeding season. Thus, for the 2026 breeding market, $G1_{j,2026}$, $G2_{j,2026}$, and $G3_{j,2026}$ summarize graded-stakes victories by stallion j's progeny during the 2025 racing year. More generally, $G1_{jt}$, $G2_{jt}$, and $G3_{jt}$ are constructed from progeny graded-stakes victories in racing year $t-1$. Racing outcomes realized during calendar year t are not used to define the recent-success variables for breeding season t.

This convention matches the market-facing information environment. Posted stud fees for a breeding season are announced before breeders allocate mares across stallions, and breeding-season promotional materials summarize information available before the season. In the stallion materials we collected for the 2026 breeding season, progeny performance information is reported through the completed 2025 racing year and does not incorporate 2026 racing results. The baseline empirical alignment therefore treats progeny graded-

stakes victories from racing year $t-1$ as predetermined public signals for breeding season t.

The price variable p_{jt} is the posted stud fee for stallion j in breeding season t, and the quantity variable q_{jt} is the number of year-t matings obtained by stallion j. The demand model uses information observed before choice to explain the year-t allocation of matings. By contrast, the future-output measures used later in the paper follow the offspring from the year-t mating cohort. These offspring outcomes are realized only after the mating choice is made and are used to test whether the public signals that shift demand also forecast subsequent elite output.

Table 1. Timing of information, choices, and outcomes

Stage	Market timing	Empirical object
Completed racing year $t-1$	Progeny racing outcomes are realized and become public before breeding season t.	Recent $G1_{jt}$, $G2_{jt}$, and $G3_{jt}$ signals; accumulated progeny information entering latent quality.
Posted fee setting	Stallion owners announce posted stud fees for the upcoming breeding season.	Posted stud fee p_{jt}.
Breeding season t	Breeders allocate mares across active stallions, with strong incentives to mate early in the season.	Mating count q_{jt} and market share s_{jt}.
Future offspring performance	Offspring from the year-t mating cohort are born, mature, and compete in later racing years.	Future elite output used in forecasting and output-alignment exercises.

Notes: The table summarizes the baseline timing used in the empirical analysis. The annual index t denotes the breeding season. Recent graded-stakes victories are constructed from the completed racing year $t-1$ and are treated as public information available before breeding-season choices in year t. Future elite output is measured from the offspring cohort generated by the year-t mating choices.

Two timing conventions are worth emphasizing. First, contemporaneous racing outcomes that occur during calendar year t are not treated as common pre-choice information for the year-t breeding market. They may affect later information sets, but not the recent-success variables used to explain year-t mating choices. Second, the empirical-Bayes latent-quality measure is aligned with the same no-look-ahead principle: information observed during year t affects later values of latent quality, but not the value used to explain year-t mating choices. These conventions prevent future or contemporaneous post-choice outcomes from mechanically entering the demand variables used to explain current choices.

2.3 Market-facing information and pedigree-based differentiation

The information environment is not limited to raw racing records. Stallions are marketed to breeders before the breeding season, and breeding-season materials present a compact summary of the information breeders are expected to use. The materials we collected typically report the posted stud fee, the stallion's own racing record, pedigree, and progeny performance. Appendix A.1 provides an illustrative example from a 2026 stallion flyer. When a stallion has recent progeny graded-stakes victories, those achievements are natural market-facing signals: they are public, easy to interpret, and directly connected to the rare outcome breeders value. The example also illustrates the timing convention used in the empirical analysis: a 2026 breeding-season material summarizes progeny performance through the completed 2025 racing year.

This market-facing role is important for interpretation. A recent graded-stakes victory by a stallion's progeny is not merely a data item observed by the econometrician. It is a public realization of elite offspring success that can enter breeders' information sets and promotional narratives before the breeding season. Such a victory may convey information about the stallion's latent ability to produce elite offspring, but it may also attract demand because it is a salient and commercially meaningful realization of the rare-success outcome itself.

Pedigree is another central dimension of market-facing information. Thoroughbred breeding decisions are shaped by sire lines, bloodline compatibility, and the commercial reputation of paternal families. Breeders may view stallions from related paternal lineages as closer alternatives because they carry related genetic and market narratives. Pedigree therefore provides a natural structure for local substitution across stallions.

The demand model operationalizes this structure using pedigree-based nests. A nest is constructed from paternal lineage: sire lines with sufficient representation among active stallions in the estimation panel are treated as separate nests, while smaller or residual lineages are pooled into an outside pedigree group. In the baseline construction, a paternal-lineage root is assigned a nest when it has at least nine direct-son stallions appearing in the estimation panel. The purpose of this rule is not to claim that pedigree fully segments the market, but to allow substitution to be more local among stallions from commercially

meaningful sire lines.

This structure will matter for interpreting substitution in the demand model. When a salient recent success signal raises the attractiveness of a focal stallion, the resulting demand gain may be drawn disproportionately from related paternal lines rather than uniformly from the whole market.

3 Data and Measurement

This section describes the data and the construction of the main empirical objects. We first describe the data sources and the posted-fee estimation sample. We then define markets, products, market shares, and the outside option. The remaining subsections construct the two main information objects used in the demand analysis: an empirical-Bayes measure of latent quality and recent graded-stakes victories as salient public success signals. The section concludes with summary statistics and basic facts about the rarity of recent elite-success signals.

3.1 Data sources and estimation sample

The empirical analysis uses a stallion-year panel for the Japanese thoroughbred breeding market from 2004 to 2024. The unit of observation is a stallion in a breeding season. This unit matches the economic object of choice because breeders choose among active stallions each season when deciding where to mate their mares. The final demand-estimation sample contains 4,467 stallion-year observations for 833 stallions.

The data combine registry-based breeding records, posted stud fees, stallion racing records, progeny racing outcomes, and pedigree information. The mating data used to construct market shares come from the Japanese Stud Book database. These records allow us to observe registered mating relationships and aggregate them to the stallion-year level. Thus, the quantity variable in the demand system is based on registered mating choices rather than on indirect sales or revenue measures.

Stallion racing records and progeny racing outcomes are based on records from the Japan Bloodstock Information System (JBIS). The working race-record files were assembled through *netkeiba* and cross-checked against JBIS. *netkeiba* reports that its racing and pedigree records are supplied under authorization from official data providers. Posted stud

fees are also collected from *netkeiba* and cross-checked against JBIS when possible. Posted fees are measured in units of ten thousand JPY and should be interpreted as publicly advertised stud-fee benchmarks rather than transaction-level realized prices.

A small number of stallions entered stud service in Japan after racing careers abroad. For these stallions, some own-racing variables, especially career race prize, are incomplete in the Japanese racing records. We supplement missing own-racing information for such stallions using Ahonoora. These supplemental records are used only for predetermined own-career covariates; mating choices, posted fees, progeny outcomes, and market-share construction remain based on the Japanese breeding and racing records described above.

Table 2 summarizes the main data inputs.

Table 2. Main data sources

Empirical object	Main source	Use in the paper
Mating records	Japanese Stud Book database	Construct stallion-year mating counts and market shares.
Outside quantity	Japanese Stud Book mare records	Construct the outside option using mares recorded as not covered in the previous year.
Posted stud fees	*netkeiba*, cross-checked against JBIS when possible	Price variable in the demand system.
Stallion own racing records	JBIS and *netkeiba*; Ahonoora for selected overseas-career missing values	Predetermined own-career controls and prior information in the empirical-Bayes quality measure.
Progeny racing outcomes	JBIS and *netkeiba*	Construct recent G1, G2, and G3 progeny success signals and future elite-output outcomes.
Pedigree links	JBIS and *netkeiba*	Construct paternal-lineage nests and pedigree-based substitution groups.

Notes: The table summarizes the main data inputs used in the empirical analysis. The working files are assembled from public Japanese racing and breeding records. Ahonoora is used only to supplement missing own-racing information for a small number of stallions with overseas racing careers before entering stud service in Japan.

The demand analysis focuses on the publicly observed posted-fee segment of the market. This restriction is necessary because posted stud fees are the price variable in the demand system. The raw fee data contain posted values, special labels, and missing observations. Observations labeled as *BOOKFUL* are excluded because a filled book indicates that capacity constraints or non-price allocation may be central to the observed mating count. These observations are therefore not treated as ordinary posted-fee choices.

The raw mating records also include stallions without usable public posted fees. We

distinguish these non-public private stallions from the posted-fee market studied in the paper. Many of these stallions have very small mating books and do not appear to be offered through the same publicly quoted stud-fee channel. We treat them as a separate fringe segment rather than as priced alternatives in the demand system. Accordingly, stallion-years without usable public or conservatively recoverable effective fees are excluded from the retained inside set. The corresponding matings are also excluded from the posted-fee market-size denominator used to construct market shares.

A small number of fee observations with special labels can be retained when an effective fee can be conservatively recovered from surrounding records. These retained observations are included only when the stallion-year can be treated as part of the publicly observed priced market. Observations with missing key covariates required for the demand specification are also excluded from the retained inside set. After applying these restrictions, the final demand-estimation sample contains 4,467 stallion-year observations for 833 stallions.

These restrictions define the empirical market studied in the paper. The retained sample should not be interpreted as a census of all Japanese breeding transactions. It corresponds to the segment of the stallion market in which mating choices, posted fees, and key covariates are observed and economically interpretable. Appendix B.1 provides additional details on special fee labels and sample construction.

3.2 Market definition and shares

A market is a breeding season indexed by year t. The demand analysis focuses on the publicly observed posted-fee segment of the Japanese stallion market. The inside products are active stallions in the retained posted-fee sample for that breeding season. Let J_t denote this retained inside set. For stallion $j \in J_t$, let q_{jt} denote the number of registered year-t matings obtained by stallion j. These mating counts are constructed from Stud Book breeding records and are the quantity measure used in the demand system.

Using mating counts rather than foal counts or later offspring outcomes is important for the economic interpretation. The breeder's choice is a mating decision made during the breeding season. Whether the mating produces a foal, whether the offspring survives and races, and whether it later becomes elite are subsequent outcomes. The demand model therefore treats registered matings as the realized allocation of breeder choices across

stallions in the posted-fee market.

The market-size normalization is defined for this posted-fee segment. Let Q_{0t} denote the number of mares recorded in the Stud Book mare records as not covered in the previous year. We use this quantity as the outside option relative to the retained posted-fee stallion market. Market size is defined as

$$M_t = \sum_{j \in J_t} q_{jt} + Q_{0t}. \tag{1}$$

The market share of stallion j in breeding season t is

$$s_{jt} = \frac{q_{jt}}{M_t}, \qquad j \in J_t, \tag{2}$$

and the outside share is

$$s_{0t} = \frac{Q_{0t}}{M_t}. \tag{3}$$

This outside option should be interpreted narrowly. It is not a complete model of every possible breeding arrangement in Japan. In particular, matings with non-public private stallions are excluded from both the inside set and the market-size denominator. These stallions do not have usable public posted fees, are not part of the publicly quoted stud-fee market, and typically have very small mating books. We therefore treat them as a separate fringe segment rather than as priced alternatives in the demand system. The resulting market definition treats the retained posted-fee stallions as the inside alternatives and mares recorded as not covered in the previous year as the outside quantity.

This normalization keeps the price and quantity objects aligned. Every inside product has a posted-fee measure and the covariates needed for demand estimation, while the outside option captures mares not covered in the relevant market-year records. The market shares used in the nested-logit demand system are therefore shares within the publicly observed posted-fee market, not shares of all possible breeding arrangements.

The resulting shares enter the nested-logit inversion through the log-share outcome

$$\log s_{jt} - \log s_{0t}. \tag{4}$$

Market-year fixed effects absorb common changes in the size and composition of the breeding market across years.

3.3 Latent quality

The central latent object in the paper is a stallion's per-mating rate of producing offspring that eventually achieve elite graded-stakes success. We refer to the empirical measure of this object as latent quality. A raw count of elite progeny is not sufficient because observed success combines two components: the stallion's underlying ability to produce elite offspring and the number of mating opportunities it receives. A stallion with more matings has more chances to produce a graded-stakes winner even if its per-mating success probability is not higher. We therefore construct an exposure-adjusted empirical-Bayes measure of latent quality.

The rare-success outcome used in this measure is a first G1, G2, or G3 victory by a stallion's offspring. Counting first victories rather than all victories gives the object a first-arrival interpretation: it measures the rate at which matings generate offspring that eventually reach elite graded-stakes success. For a mating cohort c of stallion j, let q_{jc} denote the number of matings in that cohort, and let y_{jc} denote the number of offspring from that cohort that eventually achieve their first G1, G2, or G3 victory. We model cohort-level rare-success output as

$$y_{jc} \mid \theta_j, q_{jc} \sim \text{Poisson}(q_{jc}\theta_j), \tag{5}$$

where θ_j is stallion j's latent first-graded-success rate per mating.

The empirical-Bayes construction uses predetermined public information to form an initial assessment and then updates that assessment with exposure-adjusted progeny outcomes. We place a Gamma prior on θ_j:

$$\theta_j \sim \text{Gamma}(\alpha_{j0}, \beta_{j0}), \tag{6}$$

where the Gamma distribution is parameterized by shape and rate. The prior mean is

allowed to vary with predetermined information from the stallion's own racing career:

$$\frac{\alpha_{j0}}{\beta_{j0}} = \exp(X_j'\pi). \tag{7}$$

The vector X_j contains public own-career information observed before the stallion accumulates a mature progeny record, such as career racing earnings, graded-race wins, and career starts. This prior component captures the fact that breeders and market participants can form an initial assessment of a stallion before enough offspring have raced. Appendix B.2 provides details on the estimation of the prior mean and prior strength.

The posterior update combines the prior with cohort-level first-graded-success outcomes and the mating exposure that generated them. Let $\mathcal{C}_{jt}$ denote the set of mating cohorts of stallion j whose outcome information is available before breeding season t. By Gamma-Poisson conjugacy, the posterior mean is

$$quality_{jt} = E[\theta_j \mid I_{j,t-1}] = \frac{\alpha_{j0} + \sum_{c\in\mathcal{C}_{jt}} y_{jc}}{\beta_{j0} + \sum_{c\in\mathcal{C}_{jt}} q_{jc}}. \tag{8}$$

This variable, denoted $quality_{jt}$, is the empirical-Bayes measure of latent quality used in the demand, forecasting, and pricing analyses.

The construction is aligned with the information available before breeder choice. Information observed during year t is not used to explain year-t mating choices; it can affect $quality_{j,t+1}$ and later values, but not $quality_{jt}$. The update also treats zero-success mature cohorts as informative. If a mating cohort has had enough time to reveal first graded-stakes success but produces no such offspring, its mating exposure enters the denominator of equation (8) while the numerator receives no additional success count. Young cohorts are not treated as failures before they have had sufficient time to reveal elite offspring outcomes.

This measure should be interpreted as a tractable state variable rather than as a claim that breeders literally solve the exact Bayesian updating problem in equations (5)–(8). Its role is to summarize accumulated, exposure-adjusted information about a stallion's ability to produce elite offspring. This state variable allows the empirical analysis to ask whether recent public realizations of rare success shift demand beyond what is captured by latent quality.

3.4 Recent graded-stakes success signals

Recent graded-stakes success signals are constructed separately from the latent-quality measure. For each breeding season t, the variables $G1_{jt}$, $G2_{jt}$, and $G3_{jt}$ are based on stallion j's progeny performance in the most recent completed racing year before that breeding season. Thus, for example, $G1_{j,2026}$ counts G1 victories by stallion j's progeny during the 2025 racing year. Racing outcomes realized during calendar year t are not used to define the recent-success variables for breeding season t.

The recent-signal variables are counts of graded-stakes victories, not indicators. They preserve the public hierarchy of elite racing achievement: G1 victories are the highest grade, followed by G2 and G3 victories. These counts are constructed from race results for the stallion's progeny and are aligned with the information available before breeders make year-t mating choices. They therefore enter the demand model as predetermined public signals.

The recent-signal variables differ from $quality_{jt}$ in both definition and interpretation. The latent-quality measure summarizes accumulated, exposure-adjusted information about a stallion's ability to produce offspring that eventually achieve a first G1, G2, or G3 victory. By contrast, $G1_{jt}$, $G2_{jt}$, and $G3_{jt}$ preserve newly visible, grade-specific public realizations of progeny success. These recent victories may contain information about latent quality, but they may also attract demand because they are salient, market-facing realizations of the rare-success outcome itself.

The baseline construction assigns each offspring-year to the highest grade achieved by that offspring in the relevant racing year. If an offspring wins both a lower-grade and a higher-grade race in the same year, the offspring-year contributes only to the higher-grade category. If an offspring records multiple wins within its highest achieved grade in the same year, those wins are retained. This rule prevents the same offspring-year from mechanically entering multiple grade categories while preserving the intensity of recent success within the highest grade achieved.

Because recent graded-stakes victories and latent quality are both based on public progeny performance, they are not intended to be mutually exclusive pieces of raw information. The empirical distinction is instead conceptual and econometric. Latent quality

is an exposure-adjusted state variable summarizing accumulated information about rare-success probabilities. Recent graded-stakes victories are discrete, grade-specific, and highly visible public realizations. The demand specification asks whether these recent realizations carry demand value after accounting for latent quality and other observed stallion characteristics.

3.5 Summary statistics and basic facts

Table 3 reports summary statistics for the main variables used in the analysis. The table is reported after the construction of latent quality and recent success signals because several variables are constructed objects rather than raw data fields. Panel A reports stallion-year level variables in the final posted-fee demand sample. Panel B reports time-invariant stallion-level variables after collapsing the same sample to one observation per stallion.

Table 3. Summary statistics

Variable	n	Mean	Median	SD	p10	p90
Panel A: Time-varying variables (stallion-year level)						
Posted stud fee	4,467	102.59	30.00	201.56	10.00	250.00
Mating count	4,467	50.94	23.00	58.88	2.00	143.00
Market share	4,467	0.0043	0.0020	0.0050	0.0002	0.0119
Service year	4,467	6.31	5.00	4.49	1.00	13.00
$quality_{jt}$	4,467	0.0044	0.0032	0.0040	0.0010	0.0095
$G1_{jt}$	4,467	0.08	0.00	0.52	0.00	0.00
$G2_{jt}$	4,467	0.09	0.00	0.44	0.00	0.00
$G3_{jt}$	4,467	0.16	0.00	0.67	0.00	0.00
Panel B: Time-invariant variables (stallion level)						
Career race prize	833	293.90	229.34	290.75	14.95	628.34
Career win rate	833	0.3280	0.2857	0.1954	0.1250	0.6000

Notes: Panel A reports stallion-year level variables for the final posted-fee demand sample. Panel B collapses the same sample to the stallion level by taking one observation per stallion. Posted stud fees are measured in units of ten thousand JPY. Career race prize measures total racing earnings. Mating count corresponds to current-year registered matings. Quality corresponds to the empirical-Bayes posterior mean used in demand estimation.

Several features of the sample are important for the empirical analysis. First, individual stallion shares are small. The median market share is 0.0020, and the 90th percentile is 0.0119. This reflects a differentiated market with many active stallions and motivates the use of a demand system rather than simple aggregate comparisons.

Second, posted stud fees are highly skewed. The mean posted fee is 102.59 in units of ten thousand JPY, while the median is 30.00. This skewness is consistent with substantial

heterogeneity in stallion quality, reputation, and commercial position. It also underscores the importance of accounting for quality differences and price endogeneity when estimating demand.

Third, recent graded-stakes victories are sparse. The median and 90th percentile of $G1_{jt}$, $G2_{jt}$, and $G3_{jt}$ are all zero. Only about 13.6 percent of stallion-year observations have any recent G1, G2, or G3 victory. In the current construction, about 95.5 percent of stallion-year observations have no recent G1 victory, about 94.2 percent have no recent G2 victory, and about 90.4 percent have no recent G3 victory. This sparsity is central to the paper's setting: recent elite progeny victories are rare public realizations, but the demand estimates below show that they have economically large effects when they occur.

Finally, the latent-quality measure varies meaningfully across stallion-years even though the underlying rare-success outcome is sparse. This variation comes from the empirical-Bayes update, which combines predetermined own-career information with accumulated, exposure-adjusted progeny outcomes. The summary statistics therefore illustrate the distinction used throughout the paper: latent quality is a continuous state variable summarizing accumulated rare-success information, while recent graded-stakes victories are discrete salient public signals.

4 Demand Model

This section specifies the demand system used to study breeder choice across stallions within a breeding season. The model links the two empirical objects defined above—latent quality and salient recent success signals—to the observed allocation of matings across differentiated stallion-year options. Breeders choose among stallions that differ in posted fees, latent quality, recent graded-stakes victories by their progeny, racing records, service age, and pedigree. We use a nested-logit demand system to allow substitution to be more local among stallions from related paternal lineages. We then discuss the main sources of endogeneity and the interpretation of the demand coefficients.

4.1 Nested-logit demand system

A market is a breeding season indexed by t, and a product is stallion j in that season. Let $g(j)$ denote the pedigree nest to which stallion j belongs. The mean utility from choosing

stallion j in market t is

$$\delta_{jt} = -\alpha p_{jt} + \beta_q quality_{jt} + \beta_1 G1_{jt} + \beta_2 G2_{jt} + \beta_3 G3_{jt} + W'_{jt}\gamma + \tau_t + \xi_{jt}. \tag{9}$$

Here p_{jt} is the posted stud fee, measured in units of ten thousand JPY. The variable $quality_{jt}$ is the empirical-Bayes measure of latent rare-success productivity. The variables $G1_{jt}$, $G2_{jt}$, and $G3_{jt}$ are recent graded-stakes victories by the stallion's progeny. The vector W_{jt} contains additional observed characteristics, including service year and pre-stud racing records. Market-year fixed effects τ_t absorb aggregate demand conditions and common shocks to the breeding market. The unobserved component ξ_{jt} captures stallion-year demand shifters observed by breeders but not fully observed by the econometrician.

Breeder i's indirect utility from choosing stallion j in market t is

$$u_{ijt} = \delta_{jt} + \eta_{ig(j)t} + (1-\rho)\varepsilon_{ijt}, \tag{10}$$

where $\eta_{ig(j)t}$ is a breeder-specific shock common to stallions in pedigree nest $g(j)$, and ε_{ijt} is an idiosyncratic type-I extreme-value shock. The outside option is normalized to have mean utility zero. The nesting parameter $\rho \in [0, 1)$ governs within-nest correlation in unobserved utility. When $\rho = 0$, the model reduces to the standard logit model. When $\rho > 0$, stallions in the same pedigree nest are closer substitutes than stallions in different nests.

Let s_{jt} denote stallion j's market share in year t, s_{0t} the outside share, and $s_{j|g,t}$ stallion j's share within its pedigree nest. The nested-logit inversion gives

$$\begin{aligned} \log s_{jt} - \log s_{0t} = -\alpha p_{jt} + \beta_q quality_{jt} + \beta_1 G1_{jt} + \beta_2 G2_{jt} + \beta_3 G3_{jt} \\ + W'_{jt}\gamma + \rho \log s_{j|g,t} + \tau_t + \xi_{jt}. \end{aligned} \tag{11}$$

Equation (11) is the baseline demand equation. It separates demand for latent quality from demand for recent public realizations of rare success. The pedigree nesting structure captures local substitution among related stallions while still allowing competition across the broader market.

4.2 Endogeneity and instruments

Equation (11) contains two endogenous regressors. The first is the posted stud fee p_{jt}. Stud fees are chosen by stallion owners and may reflect demand-relevant attributes observed by breeders but not fully observed by the econometrician. If high-unobserved-demand stallions also charge higher fees, treating price as exogenous would bias the estimated price sensitivity.

The second endogenous regressor is the within-nest share term $\log s_{j|g,t}$. This term is mechanically related to the allocation of demand within the pedigree nest. A positive unobserved demand shock raises both the stallion's market share and its within-nest share. Ordinary least squares would therefore confound local substitution with unobserved demand heterogeneity.

We address these sources of endogeneity using differentiation instruments constructed from rival stallion characteristics within the same market year (Berry et al., 1995; Gandhi and Houde, 2019). The instruments summarize the competitive environment faced by each stallion: how crowded or isolated it is relative to other available stallions, both within and outside its pedigree nest. The underlying rival characteristics include racing records, service age, latent quality, and recent graded-stakes success signals.

The economic logic is that rival characteristics shift a stallion's competitive position and therefore help predict posted fees and within-nest shares. At the same time, conditional on the stallion's own characteristics and market-year fixed effects, the distribution of rival characteristics is assumed not to directly enter the stallion-specific unobserved demand component ξ_{jt}. Identification therefore comes from variation across stallions within the same breeding season in their relative competitive environment.

4.3 Interpretation of demand coefficients

The parameter α measures price sensitivity in the posted-fee segment of the market. Since the price term enters utility as $-\alpha p_{jt}$, downward-sloping demand corresponds to $\alpha > 0$, or equivalently to a negative estimated coefficient on the price regressor. The estimates below use the sign and magnitude of the price coefficient to evaluate whether breeder demand becomes downward sloping after observed and latent quality differences are accounted for.

The coefficient β_q measures the demand weight placed on latent quality. Because

$quality_{jt}$ summarizes accumulated information about rare-success productivity before the breeding season, β_q captures how breeder demand responds to the empirical-Bayes state variable.

The coefficients β_1, β_2, and β_3 measure the direct demand value of recent G1, G2, and G3 progeny victories, conditional on latent quality and other observed characteristics. These coefficients are central to the paper. They capture whether salient recent realizations of rare success shift breeder demand beyond the accumulated quality state. They should not by themselves be interpreted as evidence that recent victories contain incremental information about future productivity. That question is addressed separately in the forecasting analysis below.

Finally, the nesting parameter ρ summarizes the importance of pedigree-based local substitution. A positive ρ means that unobserved utility is more correlated among stallions in the same pedigree nest, so demand lost by one stallion is more likely to be reallocated toward related stallions. This parameter links the demand system to the reallocation exercises later in the paper, where we examine where demand is drawn from when salient success signals attract additional breeder attention.

5 Demand Estimates

This section presents the main demand estimates and their economic implications. We first report how the estimated demand system changes as the specification adds richer measures of stallion quality and recent public success signals. We then translate the preferred estimates into market-share and fee-equivalent magnitudes. Finally, we use the estimated nesting structure and elasticities to describe pedigree-based substitution.

5.1 Main estimates

Table 4 reports IV nested-logit demand estimates. The specifications are organized to show how the price coefficient and the non-price demand shifters change as the information set becomes richer. Column (1) includes the posted stud fee and service year. Column (2) adds pre-stud racing records. Column (3) adds the empirical-Bayes quality measure. Column (4) is the preferred specification, which includes latent quality and recent G1, G2, and G3 progeny victories. Column (5) augments the preferred specification with additional lags of

Table 4. Nested-logit demand estimates

	(1) Baseline controls	(2) + own race records	(3) + latent quality	(4) + recent G wins	(5) + older G1 wins
Posted stud fee, p_{jt}	0.0027***	0.0021***	-0.0006	-0.0043***	-0.0041***
	(0.0004)	(0.0007)	(0.0006)	(0.0006)	(0.0008)
Service year	-0.0370***	-0.0492***	-0.0709***	-0.0940***	-0.0950***
	(0.0064)	(0.0074)	(0.0088)	(0.0108)	(0.0115)
Career race prize		0.0008***	0.0007***	0.0011***	0.0011***
		(0.0001)	(0.0001)	(0.0002)	(0.0002)
Career win rate		1.4871***	1.0876***	1.9974***	1.9315***
		(0.2325)	(0.2200)	(0.3270)	(0.3413)
$quality_{jt}$			152.1387***	153.9605***	157.9477***
			(29.6946)	(33.1614)	(33.2850)
$G1_{jt}$				0.4105***	0.3860***
				(0.0711)	(0.0844)
$G1_{j,t-2}$					-0.0124
					(0.0763)
$G1_{j,t-3}$					0.0886
					(0.0598)
$G1_{j,t-4}$					0.0724
					(0.0558)
$G2_{jt}$				0.2954***	0.2740***
				(0.0777)	(0.0798)
$G3_{jt}$				0.3624***	0.3129***
				(0.0881)	(0.0828)
Nesting parameter, ρ	0.2944***	0.2286***	0.2514***	0.1988***	0.2049***
	(0.0306)	(0.0378)	(0.0308)	(0.0347)	(0.0343)
Observations	4,467	4,467	4,467	4,467	4,467
Market-year fixed effects	Yes	Yes	Yes	Yes	Yes

Notes: Each column reports an IV nested-logit demand specification estimated on the posted-fee sample. In all columns, posted stud fee and within-nest share are treated as endogenous and instrumented using differentiation instruments constructed from rival stallion characteristics within the same market year. Column (2) adds the stallion's own racing-record variables. Column (3) adds the empirical-Bayes latent-quality measure. Column (4) adds recent G1, G2, and G3 progeny victories. Column (5) augments Column (4) with additional lags of G1 progeny victories from years $t-2$ through $t-4$. Standard errors clustered at the stallion level are reported in parentheses. Significance levels are denoted by $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

G1 victories.

The first three columns illustrate the importance of controlling for quality in this market. In the most parsimonious specifications, the coefficient on posted stud fee is positive and statistically significant. This pattern should not be interpreted as upward-sloping demand. Rather, it reflects the strong positive correlation between posted fees and omitted stallion quality. High-quality stallions charge higher fees and also attract more matings.

As the specification adds pre-stud racing records and then the empirical-Bayes quality measure, the price coefficient moves sharply toward zero. In the preferred specification in Column (4), once pre-stud reputation, latent quality, and recent graded-stakes victories are included jointly, the coefficient on posted stud fee becomes negative and statistically

significant. In the notation of Section 4, the reported price coefficient corresponds to $-\alpha$; thus, the negative coefficient in the full model is consistent with downward-sloping demand.

The non-price coefficients reveal two distinct sources of demand. The coefficient on $quality_{jt}$ is positive and precisely estimated, consistent with breeders valuing persistent information about a stallion's latent rare-success productivity. At the same time, recent G1, G2, and G3 progeny victories all enter positively and significantly. Thus, recent elite outcomes shift demand even after conditioning on accumulated latent quality, pre-stud reputation, service age, and posted fees.

Column (5) shows that this demand response is concentrated in the most recent public signal. When additional lags of G1 progeny victories are included, the coefficient on the current recent G1 signal remains large and statistically significant, while the older G1 lags are smaller and less stable. This pattern is consistent with a demand response to recent public realizations of rare success rather than a permanent premium attached to any past elite event.

The estimated nesting parameter is positive and stable across specifications. This supports the use of pedigree-based nests as an economically meaningful dimension of local substitution. Overall, Table 4 delivers three main conclusions. First, positive price coefficients in sparse specifications reflect omitted quality rather than upward-sloping demand. Second, after controlling for reputation, latent quality, and recent success signals, breeder demand is downward sloping. Third, both latent quality and salient recent success signals are important demand shifters.

5.2 Economic magnitudes

The coefficient estimates imply economically large demand responses. To interpret the magnitudes, we translate the preferred estimates into exact nested-logit market-share changes and fee-equivalent values. The market-share calculations evaluate representative stallion-years at the 25th, 50th, and 75th percentiles of the baseline market-share distribution. For each representative observation, we apply a utility shock and recompute the implied share holding all other utility components fixed.

Table 5 reports the implied effects at the median baseline share. A 0.001 increase in $quality_{jt}$ raises predicted market share by 21.1 percent, while an interquartile-range

increase raises it by 82.4 percent. A one-event increase in $G1_{jt}$ raises predicted market share by about 66.7 percent at the median representative stallion-year. The corresponding absolute increase is 0.130 percentage points. The effects of $G2_{jt}$ and $G3_{jt}$ are smaller but still economically large. A one-event increase in $G2_{jt}$ raises the median representative share by 44.4 percent, while a one-event increase in $G3_{jt}$ raises it by 57.0 percent.

Table 5. Economic magnitudes at the median baseline share

Shock	Utility shift	Relative change (%)	Share change (pp)
$quality_{jt}$, +0.001	0.154	21.1	0.041
$quality_{jt}$, IQR	0.483	82.4	0.161
$G1_{jt}$, +1	0.410	66.7	0.130
$G2_{jt}$, +1	0.295	44.4	0.087
$G3_{jt}$, +1	0.362	57.0	0.111

Notes: The table reports exact nested-logit market-share changes at the representative stallion-year closest to the median of the baseline market-share distribution. IQR denotes an interquartile-range increase in $quality_{jt}$. The relative change equals $100 \times (s'_{jt}/s_{jt} - 1)$. The share change is measured in percentage points and equals $100 \times (s'_{jt} - s_{jt})$. All calculations use the preferred demand specification and hold all other utility components fixed.

The comparison with latent quality is informative. A 0.001 increase in $quality_{jt}$ raises utility by about 0.154 and increases the median representative share by 21.1 percent, while an interquartile-range increase in $quality_{jt}$ raises utility by 0.483 and increases the median representative share by 82.4 percent. By contrast, a recent G1 progeny victory raises utility by 0.410 and increases the median representative share by 66.7 percent. A single recent G1 event is therefore much larger than a small local change in the latent-quality state, and it amounts to roughly 85 percent of the utility shift generated by an interquartile-range movement in latent quality.

A monetary translation gives a complementary scale. Using the price coefficient from the preferred specification, a utility shift Δ can be converted into a fee-equivalent value as

$$\Delta p = \frac{\Delta}{|\hat{\alpha}_p|}, \tag{12}$$

where $\hat{\alpha}_p$ denotes the estimated coefficient on the posted-fee regressor. This calculation gives the stud-fee change that would generate the same utility shift as the corresponding non-price demand shifter. The values should be interpreted as demand-side compensating-fee equivalents, not as estimates of actual fee revisions.

Table 6. Fee-equivalent interpretation of demand shifters

Variable	Change	Utility shift	Fee equivalent (10,000 JPY)
Service year	+1	-0.094	-22.0
Career race prize	98.308 to 465.754 (IQR)	0.416	97.4
Career win rate	0.2143 to 0.4667 (IQR)	0.504	117.9
$quality_{jt}$	+0.0010	0.154	36.0
$quality_{jt}$	0.0018 to 0.0049 (IQR)	0.483	112.9
$G1_{jt}$	+1	0.410	96.0
$G2_{jt}$	+1	0.295	69.1
$G3_{jt}$	+1	0.362	84.7

Notes: Fee equivalents are computed as $\Delta/|\hat{\alpha}_p|$, where $\hat{\alpha}_p$ is the estimated coefficient on posted stud fee in the preferred demand specification. IQR denotes an interquartile-range increase. Fee-equivalent values are reported in units of ten thousand JPY.

The fee-equivalent calculations reinforce the scale of the recent-success effects. A one-event increase in $G1_{jt}$ has a demand-side value of about 96.0 in units of ten thousand JPY. The corresponding values are 69.1 for $G2_{jt}$ and 84.7 for $G3_{jt}$. By comparison, a 0.001 increase in $quality_{jt}$ has a fee-equivalent value of about 36.0, while an interquartile-range increase in $quality_{jt}$ has a fee-equivalent value of 112.9. The interquartile-range movements in career race prize and career win rate have fee-equivalent values of 97.4 and 117.9, respectively. These magnitudes show that recent graded-stakes victories are not merely statistically detectable. They are large discrete demand shocks in a market where the underlying rare-success probability is small and only imperfectly observed.

5.3 Pedigree-based substitution

The nested-logit estimates also imply that pedigree is an economically meaningful dimension of substitution. In the preferred specification, the estimated nesting parameter is approximately 0.20. This value indicates a meaningful, though not overwhelming, degree of within-nest correlation in unobserved utility. Breeders appear to view stallions from related paternal lineages as closer substitutes, while the market remains broader than a set of isolated pedigree submarkets.

Additional evidence comes from the own-price elasticities implied by the full demand system. Table 7 compares observation-level own-price elasticities for stallions in pedigree nests 1–12 and for stallions in the outside group. The lineage-defined portion of the market is more price sensitive on average. The mean own-price elasticity is -0.691 for pedigree

nests 1–12, compared with -0.443 for the outside group.

Table 7. Mean own-price elasticity by aggregate pedigree grouping

Group	Mean	Median	Observations
Nests 1–12	-0.691	-0.218	1,560
Outside group	-0.443	-0.160	2,907

Notes: The table reports the mean and median of observation-level own-price elasticities implied by the preferred nested-logit demand model. The first row pools pedigree nests 1–12. The second row corresponds to the outside group.

There is also substantial heterogeneity across pedigree nests. Some lineage groups have moderate average own-price elasticities, while others are much more price sensitive. This heterogeneity is consistent with local competitive conditions differing across paternal lineages. In some nests, stallions appear to be close substitutes and therefore face stronger price pressure. In others, the available alternatives are more differentiated, and the implied price pressure is weaker.

The pedigree results are useful for interpreting the market consequences of salient recent success signals. When a recent G1, G2, or G3 victory attracts additional demand to a focal stallion, the induced demand need not come uniformly from the whole market. Part of it is drawn from nearby substitutes in the same pedigree nest, while another part comes from more distant nests and the outside option. Thus, pedigree shapes the direction of demand reallocation even though it does not fully segment the market.

Overall, the demand estimates show that breeders respond to both latent quality and salient recent success signals in a differentiated market with pedigree-based local substitution. The next section asks whether the recent signals that attract demand also contain incremental forecasting content for future elite output.

6 Salient Signals and Future Output

The demand estimates show that breeders respond strongly to recent graded-stakes victories by a stallion's progeny. This response remains sizable after controlling for latent quality, pre-stud racing records, service age, posted fees, and pedigree-based substitution. This section asks whether these salient public realizations also contain incremental information about future rare-success output. We first estimate forecasting regressions that relate recent

G1, G2, and G3 victories to future elite output from the subsequent mating cohort. We then use the estimated demand system to conduct an output-alignment exercise that varies the relative choice weight placed on latent quality and recent success signals.

6.1 Forecasting content of recent graded-stakes victories

The forecasting exercise links recent public signals to the future elite output of the mating cohort chosen after those signals are observed. For each stallion-year (j, t), let Y_{jt}^H denote the number of offspring from stallion j's year-t mating cohort whose first G1, G2, or G3 victory occurs by year $t+H$. Each offspring is counted at most once, at the time of its first graded-stakes victory. Let q_{jt} denote the number of year-t matings. The outcome therefore measures future rare-success production from the same mating cohort whose allocation is studied in the demand system.

We estimate a Poisson count model with mating exposure as an offset:

$$Y_{jt}^H \sim \text{Poisson}(\mu_{jt}^H), \tag{13}$$

where

$$\log \mu_{jt}^H = \log q_{jt} + \lambda_1 G1_{jt} + \lambda_2 G2_{jt} + \lambda_3 G3_{jt} + \phi quality_{jt} + X_{jt}'\kappa + \tau_t. \tag{14}$$

The variables $G1_{jt}$, $G2_{jt}$, and $G3_{jt}$ are the recent graded-stakes victories observed before year-t mating choices. The variable $quality_{jt}$ is the empirical-Bayes measure of latent rare-success productivity. The vector X_{jt} includes the same additional controls used in the demand specification, including service year and pre-stud racing records. The term τ_t denotes mating-year fixed effects. The offset $\log q_{jt}$ adjusts for the size of the mating cohort, so the coefficients capture associations with future elite-output rates rather than raw output counts.

A practical constraint is the maturity horizon. Future elite outcomes can only be observed once offspring have had enough time to race at high levels. At the same time, longer horizons reduce the usable sample because later mating cohorts have not fully matured. We use $H = 5$ as the baseline horizon and report $H = 6$ and $H = 7$ as maturity-window checks.

Table 8 shows that the forecasting model has meaningful predictive power for durable

Table 8. Forecasting content of recent graded-stakes victories

	(1) $H = 5$	(2) $H = 6$	(3) $H = 7$
$G1_{jt}$	-0.0202	0.0079	-0.0171
	(0.0600)	(0.0601)	(0.0652)
$G2_{jt}$	-0.0147	-0.0334	-0.0189
	(0.0428)	(0.0517)	(0.0513)
$G3_{jt}$	0.0596	0.0282	0.0434
	(0.0388)	(0.0387)	(0.0416)
$quality_{jt}$	53.2858***	56.9053***	54.5568***
	(14.4478)	(13.7373)	(14.5261)
Service year	-0.0779***	-0.0776***	-0.0755***
	(0.0184)	(0.0169)	(0.0189)
Career race prize	0.0004**	0.0004**	0.0005**
	(0.0002)	(0.0002)	(0.0002)
Career win rate	1.6617**	1.4711**	1.3695*
	(0.6966)	(0.7099)	(0.7934)
Observations	3,594	3,375	3,170
Stallions	706	677	643
Future elite events	742	889	885
Mating exposure	179,384	167,771	156,364
Mating-year range	2004–2020	2004–2019	2004–2018
Mating-year fixed effects	Yes	Yes	Yes

Notes: The dependent variable is the number of offspring from stallion j's year-t mating cohort whose first G1, G2, or G3 victory occurs by year $t+H$. All specifications are Poisson regressions with $\log(q_{jt})$ as an exposure offset, where q_{jt} is the number of year-t matings. All specifications include mating-year fixed effects. Standard errors clustered at the stallion level are reported in parentheses. Significance levels are denoted by $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

measures of stallion quality, but not for recent graded-stakes victories. The coefficient on $quality_{jt}$ is positive and statistically significant at all three horizons. Service year is negative and precisely estimated, consistent with declining future rare-success productivity over the stallion life cycle. Career race prize and career win rate are also positively associated with future elite output. Thus, the absence of robust forecasting content for recent G1, G2, and G3 signals is not driven by a general inability to predict future elite outcomes.

The recent-signal variables display a different pattern. Recent G1 and G2 victories are statistically insignificant in the baseline $H = 5$ specification and remain insignificant in the $H = 6$ and $H = 7$ checks. Recent G3 victories are positive but also statistically insignificant across the three horizons. Overall, recent G1, G2, and G3 victories provide little robust incremental forecasting content once latent quality, service age, pre-stud racing records, and year effects are held fixed.

This contrast clarifies the interpretation of the demand estimates. Breeders respond strongly to recent graded-stakes victories, and the pricing evidence below shows that the same signals are partly capitalized into posted fees. Yet these recent signals are not robust predictors of future elite output from the subsequent mating cohort after the latent-quality state is controlled for. The evidence is therefore not consistent with a simple learning interpretation in which recent public victories shift demand only because they reveal incremental productivity information. Instead, recent victories appear to operate as salient rare-success signals: they are visible and demand-relevant, but their incremental predictive content is largely absorbed by accumulated latent quality and reputation.

6.2 Output-alignment exercise

The forecasting results motivate an output-alignment exercise. The purpose is not to compute market welfare or to solve a new equilibrium. Instead, we ask a narrower question that is directly tied to breeder choice: holding other breeders' choices and the market environment fixed, would a representative breeder obtain a higher predicted probability of producing elite offspring by relying more on latent quality and less on salient recent success signals?

The exercise perturbs the representative breeder's choice weights within the estimated demand system. Prices, the product set, pedigree nests, market-year fixed effects, and all non-target utility components are held fixed. The exercise therefore does not allow stallion owners to reset fees, does not change the choices of other breeders, and does not reallocate the entire market. It is a diagnostic of whether the estimated demand weights are aligned with predicted future rare-success probabilities at the individual-choice level.

Let m_Q denote a multiplier applied to the estimated coefficient on latent quality, and let m_G denote a multiplier applied to the estimated coefficients on recent graded-stakes victories. Define the modified utility index as

$$\delta_{jt}(m_Q, m_G) = \delta^0_{jt} + m_Q \widehat{\beta}_q quality_{jt} + m_G \left(\widehat{\beta}_1 G1_{jt} + \widehat{\beta}_2 G2_{jt} + \widehat{\beta}_3 G3_{jt} \right), \tag{15}$$

where δ^0_{jt} contains all other components of the preferred demand specification. The estimated demand model corresponds to $(m_Q, m_G) = (1, 1)$.

For each market year t, the nested-logit model maps $\delta_{jt}(m_Q, m_G)$ into predicted choice probabilities. Let $\widetilde{P}_{jt}(m_Q, m_G)$ denote the representative breeder's predicted probability of choosing stallion j, conditional on choosing a stallion in the retained posted-fee inside set. This conditioning fixes the participation margin and focuses the exercise on which stallion the representative breeder would choose.

Let $\widehat{\pi}_{jt}$ denote the predicted probability that a mating with stallion j in year t produces an offspring that eventually achieves a first G1, G2, or G3 victory. We compute this probability using the forecasting specification above. The representative breeder's predicted elite-offspring probability under weights (m_Q, m_G) is

$$R_t(m_Q, m_G) = \sum_{j \in J_t} \widetilde{P}_{jt}(m_Q, m_G)\widehat{\pi}_{jt}. \tag{16}$$

Averaging across market years gives

$$R(m_Q, m_G) = \sum_t \omega_t R_t(m_Q, m_G), \tag{17}$$

where ω_t are market-year weights. We report changes relative to the estimated demand weights:

$$\Delta R(m_Q, m_G) = R(m_Q, m_G) - R(1, 1). \tag{18}$$

Positive values mean that the representative breeder would obtain a higher predicted probability of elite offspring under the alternative choice weights than under the estimated demand weights.

Figure 1 reports the output-alignment diagnostic. The figure varies the representative breeder's demand weight on latent quality and recent success signals, holding posted fees, other breeders' choices, product sets, pedigree nests, market-year fixed effects, and all other utility components fixed.

Figure 1 shows that the representative breeder's predicted probability of elite offspring increases when the choice rule places more weight on latent quality and less weight on salient recent success signals. This result is consistent with the forecasting evidence. The latent-quality measure robustly predicts future elite output, whereas recent graded-stakes victories have little robust incremental forecasting content once latent quality and other

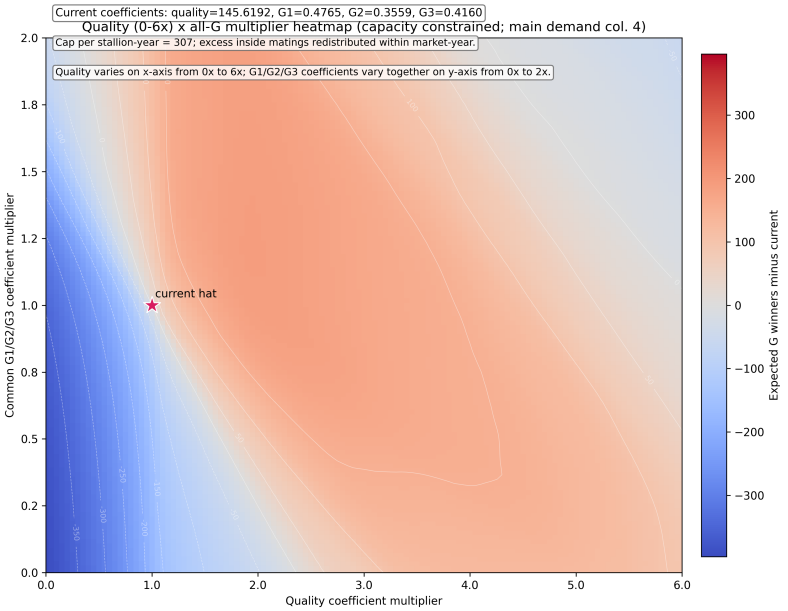


Figure 1. Output alignment for a representative breeder under alternative demand weights. The color scale reports the change in the representative breeder's predicted probability of producing elite offspring relative to the prediction under the estimated demand weights. The star marks the estimated demand weights.

durable characteristics are controlled for.

The exercise should not be interpreted as showing that breeders are irrational or that the observed market allocation is welfare inefficient. It does not model breeder heterogeneity, mare quality, private information, matching considerations, transaction costs, or stallion-owner pricing responses. Its purpose is more limited: it asks whether the demand weight attached to salient public realizations is aligned with predicted future rare-success probabilities for a representative breeder facing the observed market environment.

The answer is informative for the paper's main interpretation. Recent graded-stakes victories are public, salient, demand-relevant, and economically large demand shifters. However, when evaluated against predicted future elite-offspring probabilities, a representative breeder would obtain a higher predicted success probability by relying more on the latent-quality state and less on recent success signals. This provides the output-alignment counterpart to the demand and forecasting results: salient public realizations reshape breeder choice even when they are less aligned with future outcomes than latent quality.

7 Market Consequences

The previous sections show that recent graded-stakes victories are large demand shifters but have limited robust forecasting content beyond latent quality. This section examines two market consequences of these salient signals. First, we ask whether the same public signals are capitalized into posted stud fees. Second, we use the estimated demand system to study where the signal-induced demand gain is drawn from. The goal is not to estimate a full supply-side model or solve for a new market equilibrium. Rather, the section documents how salient rare-success realizations are reflected in prices and how they redirect demand across pedigree-structured substitutes.

7.1 Capitalization into posted fees

The demand estimates imply that recent graded-stakes victories have substantial demand value. A natural question is whether stallion owners also adjust posted fees in response to these signals. The institutional timing makes this exercise natural. Progeny racing outcomes are publicly observed before the following breeding season, stallion owners announce posted fees after this information has arrived, and breeders then make mating decisions after observing both recent signals and posted fees.

We estimate within-stallion pricing regressions of the form

$$\log(1 + p_{jt}) = \lambda_1 G1_{jt} + \lambda_2 G2_{jt} + \lambda_3 G3_{jt} + \psi quality_{jt} + Z'_{jt}\chi + \alpha_j + \tau_t + \nu_{jt}, \tag{19}$$

where p_{jt} is the posted stud fee, $quality_{jt}$ is the empirical-Bayes latent-quality measure, and $G1_{jt}$, $G2_{jt}$, and $G3_{jt}$ are recent graded-stakes victories by the stallion's progeny. The vector Z_{jt} contains additional controls such as service year. Stallion fixed effects α_j absorb time-invariant reputation, pedigree, and other persistent stallion attributes. Market-year fixed effects τ_t absorb aggregate changes in the breeding market. The coefficients on recent graded-stakes victories therefore measure whether newly observed elite progeny success is associated with higher posted fees for the same stallion.

Table 9 shows that recent elite success is selectively reflected in posted fees. Recent G1 victories are strongly associated with higher posted fees for the same stallion. The G2 and G3 coefficients are positive but not statistically significant in the baseline fixed-effects

Table 9. Capitalization into posted stud fees

	(1) $\ln(1 + \text{price}_{jt})$
$G1_{jt}$	0.1041***
	(0.0345)
$G2_{jt}$	0.0326
	(0.0273)
$G3_{jt}$	0.0156
	(0.0255)
$quality_{jt}$	73.0261***
	(16.8997)
Service year	-0.0148
	(0.1818)
Observations	4,467
Stallion fixed effects	Yes
Market-year fixed effects	Yes
Clustered standard errors	Product

Notes: The dependent variable is $\ln(1 + \text{price}_{jt})$, where price_{jt} is the posted stud fee. The specification includes stallion fixed effects and market-year fixed effects. Standard errors clustered at the product level are reported in parentheses. Significance levels are denoted by $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

specification. The coefficient on $quality_{jt}$ is positive and precisely estimated, indicating that posted fees respond not only to headline recent events but also to accumulated latent-quality information.

The evidence points to selective capitalization. Among recent success signals, the most robust price response is concentrated in recent G1 success, the most visible and commercially valuable graded-stakes signal. First-difference fee-revision specifications, reported in the appendix, show that recent G1 victories and changes in latent quality remain positively associated with annual fee revisions, while the G2 and G3 responses are less stable. We therefore interpret the pricing evidence as reduced-form capitalization rather than as a full supply-side pricing model.

These results do not imply that posted fees fully absorb the demand-side value of recent success signals. Fees may adjust only partially because of capacity constraints, long-run reputation management, contractual conventions, and frictions in year-to-year fee setting. The central message is more limited: the same salient public realizations that shift breeder demand are also reflected in the posted-fee side of the market, especially when the signal is a top-grade G1 victory.

7.2 Diversion of signal-induced demand across pedigree-structured substitutes

Recent graded-stakes victories raise demand for the focal stallion. The next question is where this signal-induced demand gain is drawn from. Because the demand model allows substitution to be more local within pedigree nests, a recent success signal may draw demand disproportionately from related stallions. At the same time, the market need not be fully segmented by pedigree; some demand may come from other nests or from the outside option.

We compute a diversion-style decomposition of the signal-induced demand gain. This object is related to diversion ratios in empirical IO, but it is not a standard price-diversion ratio. Instead of asking where demand goes when a product's price rises, we ask where the focal stallion's additional demand comes from when a recent success signal raises its utility.

For a focal stallion-year with a recent grade-k signal, where $k \in \{1, 2, 3\}$, define a counterfactual utility index that removes the direct demand contribution of that signal for the focal stallion while holding prices, product characteristics, market-year fixed effects, pedigree nests, and all other utility components fixed. Let $\widehat{s}_{jt}$ denote the predicted share under the estimated demand system, and let $\widehat{s}_{jt}^{(-k)}$ denote the predicted share after removing the grade-k recent-signal utility increment. The focal signal-induced demand gain is

$$\Delta s_{jt}^{k} = \widehat{s}_{jt} - \widehat{s}_{jt}^{(-k)}. \tag{20}$$

We decompose this gain into diversion shares from three sources. The same-nest diversion share is

$$\mathcal{D}_{jt}^{k,\text{same}} = \frac{\sum_{\ell \neq j : g(\ell) = g(j)} \left(\widehat{s}_{\ell t}^{(-k)} - \widehat{s}_{\ell t} \right)}{\Delta s_{jt}^{k}}. \tag{21}$$

The other-nest diversion share is

$$\mathcal{D}_{jt}^{k,\text{other}} = \frac{\sum_{\ell : g(\ell) \neq g(j)} \left(\widehat{s}_{\ell t}^{(-k)} - \widehat{s}_{\ell t} \right)}{\Delta s_{jt}^{k}}, \tag{22}$$

and the outside-option diversion share is

$$\mathcal{D}_{jt}^{k,\text{outside}} = \frac{\widehat{s}_{0t}^{(-k)} - \widehat{s}_{0t}}{\Delta s_{jt}^{k}}. \tag{23}$$

By construction, the three diversion shares sum to one up to numerical rounding.

Table 10. Diversion of G1-induced demand gains

Recent signal	Focal gain (pp)	Same-nest diversion (%)	Other-nest diversion (%)	Outside diversion (%)
$G1_{jt}$	0.645	39.4	53.2	7.4

Notes: The table reports a diversion-style decomposition of the G1-induced demand gain implied by the preferred nested-logit demand model. The focal gain is the average increase in the focal stallion's predicted market share from the direct G1 utility component, measured in percentage points. The diversion shares decompose this gain into demand drawn from other stallions in the same pedigree nest, stallions in other pedigree nests, and the outside option.

Table 10 shows that the G1-induced demand gain is economically meaningful and comes primarily from substitution within the retained product space. For stallion-years with a recent G1 signal, the focal stallion's predicted market share increases by about 0.645 percentage points on average. This is a large change in a market where individual product shares are small. About 39.4 percent of this gain is drawn from other stallions in the same pedigree nest, about 53.2 percent from stallions in other nests, and about 7.4 percent from the outside option.

The decomposition has two implications. First, pedigree matters for the incidence of salient success signals. A substantial share of the demand gained by a successful focal stallion is diverted from related stallions in the same paternal lineage. This is consistent with the nested-logit estimates and the elasticity evidence: pedigree structures local substitution and shapes which competitors lose demand when a salient signal raises the attractiveness of a focal stallion.

Second, pedigree does not fully segment the market. More than half of the G1-induced demand gain is diverted from outside the focal stallion's own nest, and only a small share comes from the outside option. Thus, recent elite success reshuffles demand across the market, not only within narrow lineage groups. The market response is therefore both local and broad: pedigree defines close substitutes, but salient rare-success realizations can also redirect demand across more distant substitutes.

Taken together, the pricing and diversion results show that salient public realizations have consequences beyond the focal demand coefficient. Recent success signals are reflected in posted fees, especially for G1 victories, and they redirect demand across the differentiated product space. These market consequences complement the forecasting and

output-alignment evidence. Even when recent success signals are less aligned with future outcomes than latent quality, they can reshape prices and local competitive allocation in a differentiated market for latent rare success.

8 Robustness and Scope

This section summarizes the robustness evidence reported in the main text and appendix and clarifies the scope of the interpretation. The empirical results above combine three components: an IV demand system that separates latent quality from salient recent success signals, a forecasting exercise that evaluates the incremental forecasting content of those signals, and market-consequence exercises on posted fees and signal-induced diversion. The appendix documents the measurement choices, estimation diagnostics, magnitude calculations, forecasting horizons, output-alignment diagnostics, and pricing robustness that support the main interpretation.

First, the measurement appendix documents how the posted-fee sample, latent-quality state, and recent success signals are constructed. The baseline latent-quality measure is an empirical-Bayes posterior mean that summarizes accumulated, exposure-adjusted information about a stallion's ability to produce future first G1, G2, or G3 winners. The appendix records the treatment of special fee labels, the empirical-Bayes prior, cohort alignment, zero-success mature cohorts, and the construction of recent graded-stakes success signals. These details are important because the paper's central distinction is between accumulated information about rare-success probabilities and recent public realizations of rare success.

Second, the demand appendix supports the nested-logit specification and the IV strategy. Reduced-form evidence shows that pedigree-linked recent success is associated with demand for related stallions, consistent with pedigree being an economically meaningful dimension of local substitution. The first-stage evidence shows that the excluded differentiation instruments are jointly relevant for the endogenous variables. These checks support the interpretation of the demand estimates as an IV differentiated-products demand system rather than a simple reduced-form correlation between recent success and mating counts.

Third, the forecasting results are robust in the sense most relevant for the paper's interpretation. The latent-quality measure consistently predicts future elite output from the subsequent mating cohort across the maturity horizons reported in the main text. By

contrast, recent G1 and G2 victories do not provide robust incremental forecasting content once latent quality, service age, pre-stud racing records, and year effects are controlled for. Recent G3 victories show some positive association in the shortest maturity window, but this association is weaker and does not persist across longer horizons. The main interpretation therefore does not depend on the claim that all recent signals have exactly zero predictive content. The relevant point is that recent graded-stakes victories generate large demand responses relative to their robust incremental forecasting content.

Fourth, the pricing evidence is interpreted with caution. The main-text posted-fee regression shows that recent G1 victories and latent quality are reflected in subsequent fees. Appendix pricing specifications show that recent G1 victories remain positively associated with annual fee revisions, and that the capitalization pattern is not driven by the baseline fee transformation or by the use of count variables for recent success signals. The G2 and G3 pricing responses are less stable. For this reason, we interpret the pricing results as reduced-form evidence of selective capitalization, not as a structural model of stallion-owner pricing behavior.

Fifth, the output-alignment and diversion exercises are deliberately limited. The representative breeder output-alignment exercise does not compute welfare, does not recover the utility of heterogeneous breeders, does not model mare quality, and does not solve a new equilibrium with endogenous posted fees. Instead, it asks whether, holding other breeders' choices and the market environment fixed, a representative breeder would obtain a higher predicted probability of elite offspring by placing more weight on latent quality and less weight on salient recent success signals. Appendix exercises provide supplementary diagnostics, including by-grade output-alignment figures and focal demand gains for lower-grade recent success signals. These exercises should be read as diagnostics of output alignment and demand diversion, not as welfare or general-equilibrium counterfactuals.

Several broader limitations should be kept in mind. First, the paper does not estimate a structural probability-weighting function. The comparison with insurance demand and small-probability choice motivates the measurement problem, but the empirical analysis does not identify the curvature or weighting of subjective probabilities. The paper instead estimates how demand responds to an empirical-Bayes latent-quality state and to salient public realizations of rare success.

Second, the paper does not separately identify all possible mechanisms behind the demand response to recent graded-stakes victories. Such victories may attract attention, update beliefs, increase media visibility, affect commercial narratives around a stallion, or interact with breeder preferences over pedigree and prestige. The forecasting exercise helps distinguish incremental productivity information from other demand-relevant components, but it does not isolate a single psychological mechanism. For this reason, we use the term salient success signal to describe the empirical object: a recent, public, rare, and commercially meaningful realization of the outcome breeders seek.

Third, the paper studies a specialized market. The Japanese thoroughbred stallion market is useful because choices, prices, product differentiation, rare-success outcomes, and pedigree structure are all observable in a relatively clean way. The purpose is not to claim that all rare-success markets behave like the stallion market. Rather, the setting provides a field environment in which latent rare-success probabilities and salient public realizations can be measured separately and linked to demand, posted fees, and substitution patterns.

Taken together, the robustness evidence supports the central empirical pattern. Breeder demand responds to both latent quality and salient recent success signals. The latent-quality state is more consistently aligned with future elite output, while recent graded-stakes victories generate large demand responses and are reflected in posted fees despite limited robust incremental forecasting content. These findings are most naturally interpreted as evidence that salient public realizations can reshape demand and market outcomes in differentiated markets where rare-success probabilities are latent.

9 Conclusion

Many markets require buyers to choose among differentiated products whose value depends on latent, product-specific probabilities of rare future success. In such markets, buyers must evaluate sparse information about underlying quality while also observing salient public realizations of the rare outcome they seek. This paper studies this problem in the Japanese thoroughbred stallion market, where breeders choose stallions before learning whether the resulting offspring will become elite racehorses.

The empirical analysis separates two objects that are often difficult to distinguish when

rare-success probabilities are latent. The first is latent quality, measured by an empirical-Bayes state variable that summarizes accumulated, exposure-adjusted information about a stallion's ability to produce future graded-stakes winners. The second is a salient recent success signal, measured by recent G1, G2, and G3 victories by a stallion's progeny. These recent victories are public, rare, easy to interpret, and commercially meaningful.

The main demand estimates show that breeders respond to both objects. Latent quality is strongly associated with demand, consistent with breeders valuing accumulated information about rare-success probabilities. At the same time, recent graded-stakes victories generate large demand responses even after controlling for latent quality, own racing records, service age, posted fees, and pedigree-based substitution. The economic magnitudes are large: a recent G1 victory produces a demand effect comparable to moving the stallion from the 25th to the 75th percentile of latent quality.

The forecasting results clarify the interpretation of these demand responses. Latent quality robustly predicts future elite output from the subsequent mating cohort. Recent G1 and G2 victories, by contrast, provide little robust incremental forecasting content once latent quality and other durable stallion characteristics are controlled for; the evidence for G3 is weaker and less stable across horizons. Thus, the demand response to recent success is not fully explained by a simple learning story in which recent victories merely reveal additional future productivity beyond the accumulated quality state.

The output-alignment exercise provides a complementary diagnostic. Holding other breeders' choices and the market environment fixed, a representative breeder would obtain a higher predicted probability of elite offspring by relying more on latent quality and less on salient recent success signals. This exercise is not a welfare calculation, an optimality test, or a full counterfactual equilibrium. Rather, it shows that the salient signals that strongly move demand are less aligned with predicted future rare-success outcomes than the latent-quality state.

The market consequences of salient success signals extend beyond the focal demand coefficient. Recent top-grade success is reflected in posted stud fees, especially for G1 victories, and latent quality is also reflected in fees. In addition, signal-induced demand gains are diverted from both close pedigree substitutes and more distant alternatives. Pedigree therefore shapes the local incidence of salient success signals, but it does not fully segment

the market.

Taken together, the results show that salient public realizations can reshape demand, posted fees, and demand flows across substitutes in differentiated markets where rare-success probabilities are latent. The findings do not imply that breeders are irrational, nor do they identify a structural probability-weighting function. They instead document a field pattern that is relevant for markets in which buyers value rare upside outcomes: demand may place substantial weight on vivid recent realizations even when a persistent probability state is more closely aligned with future success. This distinction between latent quality and salient public realizations is likely to matter in other settings where buyers choose products, talent, projects, or opportunities whose payoffs depend on exceptional but difficult-to-predict outcomes.

References

Ackerberg, D.A., 2001. Empirically distinguishing informative and prestige effects of advertising. RAND Journal of Economics , 316–333.

Barseghyan, L., Molinari, F., O'Donoghue, T., Teitelbaum, J.C., 2013. The nature of risk preferences: Evidence from insurance choices. American Economic Review 103, 2499–2529.

Berry, S., Levinsohn, J., Pakes, A., 1995. Automobile prices in market equilibrium. Econometrica 63, 841–890.

Berry, S.T., 1994. Estimating discrete-choice models of product differentiation. The Rand journal of economics , 242–262.

Bordalo, P., Gennaioli, N., Shleifer, A., 2012. Salience theory of choice under risk. The Quarterly Journal of Economics 127, 1243–1285.

Bordalo, P., Gennaioli, N., Shleifer, A., 2013. Salience and consumer choice. Journal of Political Economy 121, 803–843.

Chetty, R., Looney, A., Kroft, K., 2009. Salience and taxation: Theory and evidence. American economic review 99, 1145–1177.

Chiappori, P.A., Salanié, B., Salanié, F., Gandhi, A., 2019. From aggregate betting data to individual risk preferences. Econometrica 87, 1–36.

Cohen, A., Einav, L., 2007. Estimating risk preferences from deductible choice. American economic review 97, 745–788.

Crawford, G.S., Shum, M., 2005. Uncertainty and learning in pharmaceutical demand. econometrica 73, 1137–1173.

Dranove, D., Jin, G.Z., 2010. Quality disclosure and certification: Theory and practice. Journal of economic

literature 48, 935–963.

Erdem, T., Keane, M.P., 1996. Decision-making under uncertainty: Capturing dynamic brand choice processes in turbulent consumer goods markets. Marketing Science 15, 1–20.

Fay, S., Xie, J., 2008. Probabilistic goods: A creative way of selling products and services. Marketing Science 27, 674–690.

Fay, S., Xie, J., 2010. The economics of buyer uncertainty: Advance selling vs. probabilistic selling. Marketing Science 29, 1040–1057.

Gallagher, J., 2014. Learning about an infrequent event: Evidence from flood insurance take-up in the united states. American Economic Journal: Applied Economics , 206–233.

Gandhi, A., Houde, J.F., 2019. Measuring substitution patterns in differentiated-products industries. NBER Working paper .

Gandhi, A., Serrano-Padial, R., 2015. Does belief heterogeneity explain asset prices: The case of the longshot bias. The Review of Economic Studies 82, 156–186.

Grigolon, L., Verboven, F., 2014. Nested logit or random coefficients logit? a comparison of alternative discrete choice models of product differentiation. Review of Economics and Statistics 96, 916–935.

Hanyu, H., Ishii, S., Otani, S., Teramoto, K., 2025. Are final market prices sufficient for information aggregation? evidence from last-minute dynamics in parimutuel betting. arXiv preprint arXiv:2509.14645 .

Heidhues, P., Kőszegi, B., 2018. Behavioral industrial organization, in: Bernheim, B.D., DellaVigna, S., Laibson, D. (Eds.), Handbook of Behavioral Economics: Applications and Foundations 1. Elsevier North-Holland, pp. 517–612. doi:10.1016/bs.hesbe.2018.07.006.

Jin, G.Z., Leslie, P., 2003. The effect of information on product quality: Evidence from restaurant hygiene grade cards. The Quarterly Journal of Economics 118, 409–451.

Jullien, B., Salanié, B., 2000. Estimating preferences under risk: The case of racetrack bettors. Journal of Political Economy 108, 503–530.

Luca, M., Smith, J., 2013. Salience in quality disclosure: Evidence from the us news college rankings. Journal of Economics & Management Strategy 22, 58–77.

Nevo, A., 2001. Measuring market power in the ready-to-eat cereal industry. Econometrica 69, 307–342.

Ottaviani, M., Sørensen, P.N., 2009. Surprised by the parimutuel odds? American Economic Review 99, 2129–2134.

Ottaviani, M., Sørensen, P.N., 2010. Noise, information, and the favorite-longshot bias in parimutuel predictions. American Economic Journal: Microeconomics 2, 58–85.

Sydnor, J., 2010. (over) insuring modest risks. American Economic Journal: Applied Economics 2, 177–199.

Thaler, R.H., Ziemba, W.T., 1988. Anomalies: Parimutuel betting markets: Racetracks and lotteries. Journal of Economic perspectives 2, 161–174.

A Institutional Details

A.1 Illustrative stallion breeding-season materials

This appendix provides an illustrative example of the market-facing materials discussed in Section 2.3. The example is a 2026 breeding-season material for Real Steel. The purpose of the exhibit is not to document every stallion flyer or to construct the estimation variables from promotional materials. Rather, it illustrates the type of information that is presented to breeders before the breeding season and helps motivate why recent progeny graded-stakes victories are natural salient public signals in this market.

Figure 2 shows the first spread of the material. The page displays the stallion's identity, photograph, birth year, pedigree, body measurements, own racing record, and the 2026 posted stud fee. The posted fee is displayed as a breeding-season term, and the page also reports booking status. Thus, the material presents the main objects that enter breeders' choice environment: the stallion's price, own racing credentials, pedigree, and market positioning for the coming breeding season.

Figure 3 shows the subsequent spread. This page summarizes the stallion's stud record, recent progeny performance, paternal line, and female family. In particular, the material highlights recent progeny race achievements, including graded-stakes victories by offspring. This is the market-facing object corresponding to the recent success signals used in the demand analysis. The example also illustrates the timing convention used in the paper: the 2026 breeding-season material summarizes progeny performance through the completed 2025 racing year rather than incorporating 2026 racing results.

The example supports two institutional points used in the main text. First, recent progeny graded-stakes victories are not merely variables constructed by the econometrician after the fact. They are public race outcomes that can appear in the market-facing information environment in which breeders evaluate stallions. Second, these achievements are presented alongside posted fees, own racing records, and pedigree information. This supports the empirical distinction between accumulated latent quality and recent salient public realizations of rare success: both are visible to the market, but recent progeny victories are discrete, grade-specific, and especially easy to interpret.



Figure 2. Example of breeding-season stallion material: price, own racing record, and pedigree



2025年、レーベンスティールとアドマイヤクワッズはG２優勝

種牡馬成績 Stud Record

2019年から供用。2024年より当場にて供用開始。総合ファーストシーズンサイアーランキング2位(2022年)。

【主な産駒】

フォーエバーヤング(21 牡 BMS:Congrats)年度代表馬, 最優秀古牡馬, 最優秀ダートホース, ＪＲA特別賞, ＮＡＲダートグレード競走特別賞馬, ＢＣクラシック-**G1**, 同３着, サウジC-**G1**, 東京大賞典-**G1**, ジャパンダートクラシック-**JPN1**, 全日本２歳優駿-**JPN1**, ＵＡＥダービー-**G2**, 日本テレビ盃-**JPN2**, サウジダービー-**G3**, ＪＢＣ２歳優駿-**JPN3**, ケンタッキーダービー-**G1**３着, ドバイワールドC-**G1**３着

レーベンスティール(20 牡 BMS:トウカイテイオー)毎日王冠-**G2**, セントライト記念-**G2**, オールカマー-**G2**, エプソムC-**G3**

アドマイヤクワッズ(23 牡 BMS:Zoffany)デイリー杯２歳Ｓ-**G2**, 朝日杯フューチュリティＳ-**G1**３着

オールパルフェ(20 牡 BMS:ルーラーシップ)デイリー杯２歳Ｓ-**G2**

チカッパ(21 牡 BMS:Into Mischief)東京盃-**JPN2**, 北海道スプリントC-**JPN3**, 昇竜Ｓ-**OP**, ＪＢＣスプリント-**JPN1**２着

ヴェローチェエラ(21 牝 BMS:Danehill Dancer)函館記念-**G3**, 京都新聞杯-**G2**３着

カナルビーグル(22 牡 BMS:Include)ユニコーンＳ-**G3**, 東京ダービー-**Jpn1**５着

フェブランシェ(20 牝 BMS:*クロフネ)スパーキングレディーC-**JPN3**

ノイジーボーイ Noisy Boy (20 騙 BMS:Northern Meteor)ＡＴＣクラックＳ-**G2**２着

ヴィンセンシオ(22 牡 BMS:キングカメハメハ)業牡丹賞, 弥生賞ディープインパクト記念-**G2**２着

ドナベティ(21 牝 BMS:Malibu Moon)すずらん賞-**OP**, ファンタジーＳ-**G3**２着

ティラトーレ(22 牝 BMS:ルーラーシップ)フェアリーＳ-**G3**２着, クロッカスＳ-**L**２着

ショウナンカリス(23 牝 BMS:*アメリカンファラオ)ファンタジーＳ-**G3**２着

2025 ＢＣクラシック-**G1** (Del Mar・D10F)

2025 毎日王冠-**G2** (東京・芝1800m)

2025 デイリー杯２歳Ｓ-**G2** (京都・芝1600m)

アレンジャー(21 牡 BMS:ネオユニヴァース)アーリントンC-**G3**２着

タイセイプランセス(22 牝 BMS:Australia)フローラＳ-**G2**３着

ビターヤ(21 牡 BMS:Singspiel)欅Ｓ-**OP**, 東海Ｓ-**G3**３着, 武蔵野Ｓ-**G3**３着

ナムラエイハブ(21 牡 BMS:*エンパイアメーカー)弥彦Ｓ, 栗島特別, カシオペアＳ-**L**２着

80

父は本邦＆２歳チャンピオンサイアー

父系 Sire

父:**ディープインパクト**は早来産。２～４歳時に日仏で14戦12勝。年度代表馬・２回, 最優秀３歳牡馬, 最優秀古牡馬, ジャパンC-**G1**, 三冠。総合チャンピオンサイアー(2012～2022年), ２歳チャンピオンサイアー(2010～2014年, 2016～2021年)。代表産駒に

リアルスティール本馬

オーギュストロダン Auguste Rodin (20 牡 BMS:Galileo)英ダービー-**G1**, 愛ダービー-**G1**, 愛チャンピオンＳ-**G1**, ＢＣターフ-**G1**, プリンスオブウェールズＳ-**G1**

ジェンティルドンナ(09 牝 BMS:Bertolini)年度代表馬・２回, 最優秀３歳牝馬, 最優秀古牝馬・２回, ジャパンC-**G1**・２回, 有馬記念-**G1**

全妹ラヴズオンリーユーは日米・香港の３カ国でＧ１勝利

母系 Family

母 *ラヴズオンリーミー Loves Only Me は米国産,不出走。産駒

ラッドルチェンド(10 牝 鹿 Danehill Dancer)入着,南関東(公)1勝。産駒

テルツェット(牝 ディープインパクト)7勝,ダービー卿チャレンジT-**G3**,クイーンＳ-**G3**(2回),中山牝馬Ｓ-**G3**５着

ノクターン(牝 キタサンブラック)1勝,忘れな草賞-**L**３着,㊙

ラングレー(11 牡 鹿 ディープインパクト)6勝,洛陽Ｓ-**OP**,石清水Ｓ,初富士Ｓ,ゆりかもめ賞,都大路Ｓ-**L**３着,毎日杯-**G3**４着。種牡馬

リアルスティール(12 牡 鹿 ディープインパクト)**本馬**

プロディガルサン(13 牡 鹿 ディープインパクト)3勝,芙蓉Ｓ-**L**,ロードカナロアメモリアル,東京新聞杯-**G3**２着,東京スポーツ杯２歳Ｓ-**G3**２着,六甲Ｓ-**L**２着,ディセンバーＳ-**L**２着,谷川岳Ｓ-**L**２着,セントライト記念-**G2**３着

ジンゴイスト(15 騙 青鹿 ハーツクライ)東海(公)2勝,中央1勝,障2勝

ラヴズオンリーユー(16 牝 鹿 ディープインパクト)最優秀古牝馬,5勝,オークス-**G1**,京都記念-**G2**,忘れな草賞-**L**,白菊賞,札幌記念-**G2**２着,鳴尾記念-**G3**２着,エリザベス女王杯-**G1**３着(２回),府中牝馬Ｓ-**G2**５着,香港２勝,香港C-**G1**,香港クイーンエリザベス二世C-**G1**,米芝牝馬チャンピオン,北米1勝,ＢＣフィリー&メアターフ-**G1**,UAE入着,ドバイシーマクラシック-**G1**３着

ラヴユーライヴ(17 牝 鹿 ディープインパクト)4勝,立雲峡Ｓ,支笏湖特別,ケフェウスＳ-**OP**３着

ディーンズリスター(19 騙 鹿 ディープインパクト)1勝,㊙

バトルハーデン(20 牡 鹿 ドゥラメンテ)3勝,葉山特別。種牡馬

グラヴィス(21 牡 鹿 ハーツクライ)3勝,日進特別

祖母モネヴァッシア Monevassia は仏入着。**ランブルスティルツキン** Rumplestiltskin (全欧2歳牝馬チャンピオン,モイグレアスタッドS-**G1**),**アイアムビューティフル** I am Beautiful (バランシーンS-**G3**)の母,**タペストリー** Tapestry (ヨークシャーオークス-**G1**),**ジョンエフケネディ** John F Kennedy (ジュヴェナイルターフS-**G3**)の祖母

リアルスティール

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REAL STEEL (JPN)

Figure 3. Example of breeding-season stallion material: progeny performance and pedigree lines

B Data Construction and Measurement Details

This appendix provides additional details on the measurement choices underlying the posted-fee sample, the empirical-Bayes latent-quality measure, and the recent graded-stakes success variables. The main text defines the core empirical objects. This appendix records the implementation details that support those definitions.

B.1 Posted-fee sample and special fee labels

The demand analysis focuses on the publicly quoted posted-fee segment of the Japanese thoroughbred stallion market. This restriction is imposed because posted fees are the price variable in the demand system. The raw fee data include posted values, special labels, and missing observations. We retain stallion-year observations only when the posted fee is usable or can be conservatively recovered.

Observations labeled as BOOKFUL are excluded from the retained inside set. A BOOKFUL label indicates that the stallion's breeding book was filled before a standard posted fee can be treated as the relevant market price. In such cases, realized matings may reflect capacity rationing, relationship-based allocation, or other non-price mechanisms rather than ordinary posted-fee choice.

Observations labeled as PRIVATE are treated differently. These observations are retained when an effective fee can be recovered from nearby observed values in a conservative way. When no usable fee can be recovered, the observation is excluded. Observations with missing key covariates required for the demand specification are also excluded from the retained inside set.

These sample restrictions imply that the demand estimates should be interpreted as describing breeder choice within the observable posted-fee segment of the market. They should not be read as a census of all Japanese breeding transactions.

B.2 Empirical-Bayes prior construction

The main text defines $quality_{jt}$ as an empirical-Bayes posterior mean for stallion j's latent rare-success productivity before breeding season t. This subsection provides the full prior construction.

Let θ_j denote stallion j's latent first-graded-success productivity per mating. For a

mating cohort c, let q_{jc} denote mating exposure and let y_{jc} denote the number of offspring from that cohort that eventually achieve their first G1, G2, or G3 victory. The cohort-level rare-success model is

$$y_{jc} \mid \theta_j, q_{jc} \sim \text{Poisson}(q_{jc}\theta_j). \tag{24}$$

We use a Gamma prior,

$$\theta_j \sim \text{Gamma}(\alpha_{j0}, \beta_{j0}), \tag{25}$$

where the Gamma distribution is parameterized by shape and rate. The prior mean is α_{j0}/β_{j0}.

The prior mean is allowed to vary with predetermined information from the stallion's own racing career:

$$\frac{\alpha_{j0}}{\beta_{j0}} = \exp(X_j'\pi). \tag{26}$$

The vector X_j includes publicly visible pre-progeny information such as log career racing earnings, career G1 wins, career G2 wins, career G3 wins, and total career starts. These variables are fixed before the stallion's progeny record becomes informative and summarize the racing credentials available for initial assessments.

The prior strength is governed by a common shrinkage parameter β_0. We set

$$\beta_{j0} = \beta_0 \qquad \text{and} \qquad \alpha_{j0} = \beta_0 \exp(X_j'\pi). \tag{27}$$

The coefficients π are estimated from mature cohort-level first-graded-success output using a Poisson model with mating exposure. The parameter β_0 is estimated using the Gamma-Poisson marginal likelihood. A larger β_0 assigns more prior exposure to the predicted prior mean, while a smaller β_0 allows realized progeny outcomes to move the posterior more quickly.

B.3 Cohort alignment and posterior updating

A measurement issue is that mating exposure and offspring performance are observed at different points in time. The mating year determines the exposure cohort, while the first graded-stakes victory may occur several years later. We therefore align each first-graded-success outcome to the mating cohort from which the offspring was produced. This cohort

alignment avoids imposing a single fixed lag between mating exposure and observed progeny success.

Let $\mathcal{C}_{jt}$ denote the set of mating cohorts of stallion j whose outcome information is available before breeding season t. For each cohort in $\mathcal{C}_{jt}$, the update uses both the number of first-graded-success offspring and the corresponding mating exposure. By Gamma-Poisson conjugacy, the posterior mean is

$$quality_{jt} = E[\theta_j \mid I_{j,t-1}] = \frac{\alpha_{j0} + \sum_{c \in \mathcal{C}_{jt}} y_{jc}}{\beta_{j0} + \sum_{c \in \mathcal{C}_{jt}} q_{jc}}. \tag{28}$$

The set $\mathcal{C}_{jt}$ includes successful mature cohorts and mature cohorts with no first-graded-success offspring. Zero-success mature cohorts are informative because they increase the denominator without increasing the numerator. Excluding these cohorts would mechanically overstate rare-success productivity. Young cohorts are not treated as failures before they have had sufficient time to reveal elite offspring outcomes.

The posterior is aligned with the breeder's information set. Information observed during year t affects $quality_{j,t+1}$ and later values, but not $quality_{jt}$. When historical progeny outcomes and mating exposure are available before the estimation panel begins, the posterior incorporates this pre-sample information.

The posterior mean is used as a parsimonious empirical-Bayes state variable. It should not be interpreted as a claim that breeders literally solve the exact updating problem estimated here. Its role is to summarize accumulated, exposure-adjusted information about rare-success productivity at the time breeders evaluate stallion j in season t.

B.4 Recent graded-stakes success signals

Recent graded-stakes success variables are constructed separately from the latent-quality measure. For each stallion-year, $G1_{jt}$, $G2_{jt}$, and $G3_{jt}$ are constructed from the previous racing year's progeny performance. These variables count recent progeny wins in the corresponding graded race categories and are predetermined public signals when breeders make year-t mating decisions.

The quality measure and the recent-signal variables differ in both timing and interpretation. The quality measure uses accumulated first-graded-success outcomes and mating

exposure to summarize persistent productivity per mating. The recent $G1_{jt}$, $G2_{jt}$, and $G3_{jt}$ variables preserve the grade of newly visible success and allow recent high-salience events to enter separately from the slow-moving posterior state.

To avoid overlap across grades, the baseline construction assigns each offspring-year to the highest grade achieved by that offspring in the calendar year. If an offspring wins both a lower-grade and a higher-grade race in the same year, the offspring-year contributes only to the higher-grade measure. If an offspring records multiple wins within its highest achieved grade in the same year, those wins are retained. This rule preserves the intensity of recent graded success while preventing the same offspring-year from mechanically entering multiple grade categories.

This construction makes the empirical distinction in the main text operational: $quality_{jt}$ captures accumulated exposure-adjusted evidence about latent rare-success productivity, whereas $G1_{jt}$, $G2_{jt}$, and $G3_{jt}$ capture newly visible public realizations of rare success.

C Demand Estimation Details

This appendix provides additional details on the empirical support for the nested-logit demand specification. The main text presents the demand model, the endogeneity concerns, and the preferred estimates. This appendix adds two supporting pieces: reduced-form evidence that pedigree groups capture meaningful demand comovement, and first-stage evidence for the excluded differentiation instruments.

C.1 Pedigree-linked demand comovement

The main text uses pedigree-based nests to allow substitution to be more local among stallions from related paternal lineages. This subsection provides reduced-form evidence that such lineage groups capture meaningful market comovement. The exercise is not intended to estimate substitution structurally. Its purpose is to show that paternal-lineage groups form economically relevant comparison sets in breeder choice.

We estimate the following within-stallion specification:

$$\begin{aligned}\log q_{jt} = {} & \delta_1 G1_{jt} + \delta_2 G2_{jt} + \delta_3 G3_{jt} + \phi_1 G1_{jt}^{same} + \phi_2 G2_{jt}^{same} + \phi_3 G3_{jt}^{same} \\ & + \gamma quality_{jt} + \lambda serviceyear_{jt} + \mu_j + \tau_t + u_{jt}.\end{aligned} \tag{29}$$

Here q_{jt} is the mating count of stallion j in breeding season t. The variables $G1_{jt}$, $G2_{jt}$, and $G3_{jt}$ measure recent graded-stakes victories by the focal stallion's progeny. The variables $G1_{jt}^{same}$, $G2_{jt}^{same}$, and $G3_{jt}^{same}$ measure the corresponding recent victories produced by other stallions in the same pedigree nest and market year. The specification includes stallion fixed effects μ_j and market-year fixed effects τ_t.

Table 11. Reduced-form evidence for pedigree-linked demand comovement

	$\log q_{jt}$
$G1_{jt}$	0.0871***
	(0.0324)
$G2_{jt}$	0.0566
	(0.0374)
$G3_{jt}$	0.0570*
	(0.0323)
$G1_{jt}^{same}$	0.0266***
	(0.0100)
$G2_{jt}^{same}$	-0.0014
	(0.0064)
$G3_{jt}^{same}$	-0.0051
	(0.0061)
$quality_{jt}$	38.9223**
	(15.3437)
Service year	0.0462
	(0.1747)
Observations	4,331
Stallion fixed effects	Yes
Market-year fixed effects	Yes
Clustered standard errors	Stallion

Notes: The dependent variable is the log mating count. The variables $G1_{jt}$, $G2_{jt}$, and $G3_{jt}$ measure recent graded-stakes victories by the focal stallion's progeny. The variables $G1_{jt}^{same}$, $G2_{jt}^{same}$, and $G3_{jt}^{same}$ measure corresponding victories produced by other stallions in the same pedigree nest and market year. The regression includes stallion and market-year fixed effects. Standard errors clustered at the stallion level are reported in parentheses. Significance levels are denoted by $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 11 shows that recent G1 success by the focal stallion is positively associated with its mating count. More importantly for the nesting structure, recent G1 success by other stallions in the same pedigree nest is also positive and statistically significant. This pattern indicates that elite progeny success within a paternal lineage is associated with higher demand for related stallions, even after conditioning on the focal stallion's own recent signals, latent quality, service year, stallion fixed effects, and market-year fixed effects. The evidence supports the interpretation that pedigree groups capture a relevant

dimension of market comparison.

C.2 Instrument construction and first-stage relevance

The preferred demand specification treats posted stud fees and within-nest shares as endogenous. The excluded instruments are differentiation instruments constructed from rival stallion characteristics within the same market year. These instruments summarize how crowded or isolated stallion j is relative to other available stallions. Some instruments measure market-wide differentiation, while others measure crowding within and outside the stallion's pedigree nest.

The underlying rival characteristics include pre-stud racing records, service year, latent quality, and recent graded-stakes success signals. The identifying logic is that rival characteristics shift the competitive environment faced by a stallion and therefore help predict posted fees and within-nest shares. Conditional on the stallion's own observed characteristics and market-year fixed effects, the distribution of rival characteristics is assumed not to directly enter the stallion-specific unobserved demand component.

Table 12. First-stage relevance summary

Statistic	Value
Number of excluded instruments	19
Wald χ^2 for excluded instruments	133.19
p-value	< 0.001
R^2	0.675
Observations	4,467

Notes: The table summarizes first-stage relevance for the baseline excluded-instrument set in the posted-fee first stage. The Wald statistic reports the heteroskedasticity-robust joint test that all excluded-instrument coefficients are zero.

Table 12 shows that the excluded instruments are jointly relevant after conditioning on own characteristics and market-year fixed effects. The IV demand estimates in the main text should therefore be interpreted as using rival-characteristics variation to recover price sensitivity and pedigree-based local substitution in the nested-logit demand system.

D Economic Magnitudes and Elasticity Calculations

This appendix documents how the economic magnitudes and elasticity objects are computed from the preferred nested-logit demand specification. The main text reports the

median representative market-share effects, fee-equivalent values, and aggregate own-price elasticity comparison. This appendix provides the exact share-response formulas, full representative-share tables, observation-level share-gain distribution, and cross-price elasticity construction.

D.1 Exact nested-logit share calculations

The market-share effects in the main text are evaluated for representative stallion-years defined by the 25th, 50th, and 75th percentiles of the baseline market-share distribution. For each representative percentile $r \in \{25, 50, 75\}$, let q_r denote the corresponding percentile of the observed product-share distribution. We select the observed stallion-year whose baseline market share is closest to q_r and denote its product share and within-nest share by $\left(s^{(r)}, s_{j|g}^{(r)}\right)$. Let

$$S_g^{(r)} = \frac{s^{(r)}}{s_{j|g}^{(r)}} \tag{30}$$

denote the implied group share for representative percentile r.

Consider a utility shift Δ for stallion j within nest g. In the nested-logit model, this shock affects both the stallion's within-nest share and the total share of the nest. For representative percentile r, define

$$M^{(r)} = 1 - s_{j|g}^{(r)} + s_{j|g}^{(r)} \exp\left(\frac{\Delta}{1-\rho}\right). \tag{31}$$

The updated group share is

$$S_g'^{(r)} = \frac{S_g^{(r)} \left(M^{(r)}\right)^{1-\rho}}{1 - S_g^{(r)} + S_g^{(r)} \left(M^{(r)}\right)^{1-\rho}}, \tag{32}$$

and the updated within-nest share is

$$s_{j|g}'^{(r)} = \frac{s_{j|g}^{(r)} \exp\left(\frac{\Delta}{1-\rho}\right)}{M^{(r)}}. \tag{33}$$

The updated product share is therefore

$$s'^{(r)} = S_g'^{(r)} s_{j|g}'^{(r)}. \tag{34}$$

We report both the relative and absolute share changes:

$$\text{Relative change}^{(r)} = 100 \times \left(\frac{s'^{(r)}}{s^{(r)}} - 1 \right), \tag{35}$$

and

$$\text{Percentage-point change}^{(r)} = 100 \times \left(s'^{(r)} - s^{(r)} \right). \tag{36}$$

For recent graded-stakes outcomes, the utility shift is the corresponding estimated coefficient. For example, the market-share effect of a recent G1 victory is computed by setting $\Delta = \widehat{\beta}_{G1}$. For the latent-quality measure, a fixed increase of 0.001 is computed by setting $\Delta = \widehat{\beta}_q \times 0.001$.

D.2 Representative-share effects

Table 13 reports exact nested-logit share calculations for one-event increases in recent G1, G2, and G3 progeny victories at representative baseline shares.

Table 13. Demand effects of recent elite-offspring events at representative baseline market shares

Event	Share pctile	Baseline share (%)	Relative change (%)	Share change (pp)	Utility shift	Zero-share (%)
$G1_{jt}$	P25	0.052	80.8	0.042	0.476	95.5
$G1_{jt}$	P50	0.195	80.6	0.157	0.476	95.5
$G1_{jt}$	P75	0.695	79.7	0.554	0.476	95.5
$G2_{jt}$	P25	0.052	55.6	0.029	0.356	94.2
$G2_{jt}$	P50	0.195	55.5	0.108	0.356	94.2
$G2_{jt}$	P75	0.695	55.0	0.382	0.356	94.2
$G3_{jt}$	P25	0.052	67.7	0.035	0.416	90.4
$G3_{jt}$	P50	0.195	67.5	0.132	0.416	90.4
$G3_{jt}$	P75	0.695	66.9	0.465	0.416	90.4

Notes: Representative stallion-years are the observed stallion-year observations closest to the 25th, 50th, and 75th percentiles of the baseline market-share distribution. Relative change reports $100 \times (s'/s - 1)$, and share change (pp) reports $100 \times (s' - s)$ using the exact nested-logit counterfactual. Zero-share reports the share of stallion-year observations with no lagged win of the corresponding grade.

Table 14 reports the corresponding calculations for a 0.001 increase in the latent-quality measure.

Table 14. Demand effects of a 0.001 increase in $quality_{jt}$ at representative baseline market shares

Share pctile	Baseline share (%)	Quality change	Utility shift	Relative change (%)	Share change (pp)
P25	0.052	0.0010	0.146	19.8	0.010
P50	0.195	0.0010	0.146	19.8	0.039
P75	0.695	0.0010	0.146	19.7	0.137

Notes: $quality_{jt}$ is the posterior mean of expected breeding effectiveness per mating. The quality shock is a fixed increase of 0.001. Representative stallion-years are defined as the observed stallion-year observations closest to the 25th, 50th, and 75th percentiles of the baseline market-share distribution. Relative change reports $100 \times (s'/s - 1)$, and share change (pp) reports $100 \times (s' - s)$ using the exact nested-logit counterfactual.

These tables extend the median-share effects reported in the main text by showing the same calculations at the 25th and 75th percentiles of the baseline share distribution. They show that proportional effects are similar across representative shares, while absolute percentage-point gains are larger for stallions that already occupy larger market positions.

D.3 Observation-level share-gain distribution

The representative-share calculations summarize effects at three baseline market positions. Table 15 instead computes the exact nested-logit share response for every stallion-year in the estimation sample and summarizes the distribution of predicted gains.

Table 15. Observation-level distribution of predicted market-share gains

Shock	Mean gain (pp)	Median gain (pp)	p25	p75	Mean relative (%)	Median relative (%)
$G1_{jt}$	0.330	0.154	0.042	0.540	79.2	80.4
$G2_{jt}$	0.228	0.106	0.029	0.373	54.7	55.4
$G3_{jt}$	0.277	0.129	0.035	0.454	66.4	67.4
$quality_{jt}$ IQR	0.349	0.163	0.045	0.572	84.0	85.4

Notes: This table reports observation-level predicted market-share gains implied by the full nested-logit demand specification. For each stallion-year observation in the estimation sample, the table applies a counterfactual utility shock and computes the exact nested-logit share response holding all other characteristics fixed. $G1_{jt}$, $G2_{jt}$, and $G3_{jt}$ shocks correspond to one-unit increases in the respective recent offspring-performance variables. The $quality_{jt}$ IQR shock corresponds to an increase in the $quality_{jt}$ measure from the 25th to the 75th percentile of its sample distribution. Reported share gains are measured in percentage points, where share gain (pp) equals $100 \times (s' - s)$ and relative change equals $100 \times (s'/s - 1)$.

This distributional summary complements the main-text magnitudes. It shows how the same utility shocks translate into different share gains depending on baseline product

shares, within-nest shares, and market composition.

D.4 Fee-equivalent convention

The main text reports the fee-equivalent interpretation of the main demand shifters. We do not reproduce that table here. The convention is that a utility shift Δ is converted into a compensating fee change using

$$\Delta p = \frac{\Delta}{|\widehat{\alpha}_p|}, \tag{37}$$

where $\widehat{\alpha}_p$ is the estimated coefficient on the posted-fee regressor in the preferred demand specification. These values should be interpreted as demand-side fee equivalents, not as actual fee revisions or supply-side pass-through estimates.

D.5 Own-price and cross-price elasticities

Own-price elasticities are computed at the observation level and then averaged across the sample. For each stallion-year observation, the own-price elasticity implied by the nested-logit model is

$$\varepsilon_{jj,t} = -\alpha p_{jt} \left(\frac{1}{1-\rho} - \frac{\rho}{1-\rho} s_{j|g,t} - s_{jt} \right), \tag{38}$$

where p_{jt} is the posted stud fee, s_{jt} is the product share, and $s_{j|g,t}$ is the within-nest share for stallion j in year t. The main text reports the aggregate comparison between pedigree nests 1–12 and the outside group. Table 16 reports the corresponding observation-level own-price elasticities separately for each pedigree nest.

Table 16. Mean observation-level own-price elasticity by pedigree nest

Nest	Mean	Median	Obs.
Nest 1	-0.544	-0.172	670
Nest 2	-1.316	-0.457	236
Nest 3	-1.067	-0.286	137
Nest 4	-0.340	-0.167	132
Nest 5	-0.373	-0.171	50
Nest 6	-1.006	-0.654	79
Nest 7	-0.619	-0.413	65
Nest 8	-0.486	-0.287	53
Nest 9	-0.521	-0.158	40
Nest 10	-0.482	-0.283	39
Nest 11	-0.417	-0.130	30
Nest 12	-2.774	-2.257	29
Nest 0 (Outside)	-0.478	-0.173	2,907

Notes: Each row reports the mean and median of observation-level own-price elasticities implied by the nested-logit model within a pedigree nest. Nest 0 is the outside group.

To study pedigree-based substitution more directly, we also compute pairwise cross-price elasticities implied by the preferred nested-logit specification. For a given year t, the cross-price elasticity of demand for stallion j with respect to the price of stallion k is

$$\varepsilon_{jkt} = -\alpha p_{kt} s_{kt}, \qquad \text{if } j \text{ and } k \text{ are in different nests,} \tag{39}$$

and

$$\varepsilon_{jkt} = -\alpha p_{kt} \left(s_{kt} + \frac{\rho}{1-\rho} s_{k|g,t} \right), \qquad \text{if } j \text{ and } k \text{ are in the same nest.} \tag{40}$$

Here p_{kt} is stallion k's posted stud fee, s_{kt} is stallion k's market share, and $s_{k|g,t}$ is stallion k's within-nest share in year t.

For each pair (j, k), we average year-specific cross-price elasticities across the set of years in which both stallions are simultaneously present:

$$\bar{\varepsilon}_{jk} = \frac{1}{T_{jk}} \sum_{t \in \mathcal{T}_{jk}} \varepsilon_{jkt}, \tag{41}$$

where $\mathcal{T}_{jk}$ denotes the set of overlapping years and $T_{jk} = |\mathcal{T}_{jk}|$. To construct the heatmap,

we symmetrize pairwise elasticities by averaging the two directional objects:

$$\bar{\varepsilon}_{jk}^{sym} = \frac{\bar{\varepsilon}_{jk} + \bar{\varepsilon}_{kj}}{2}. \tag{42}$$

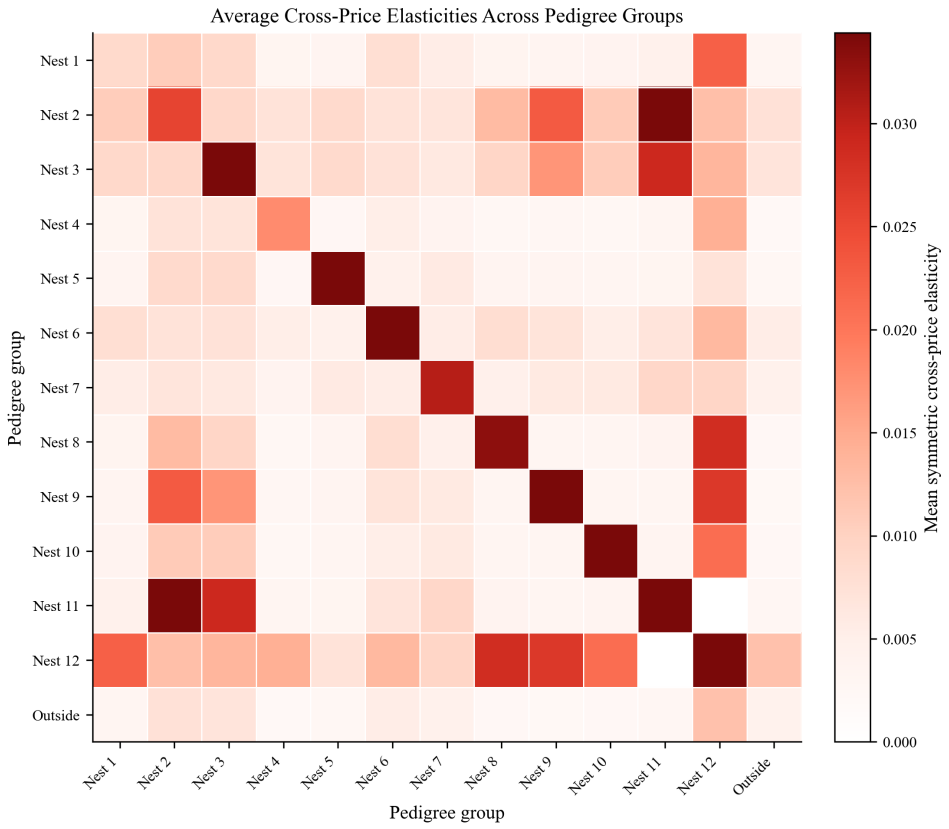


Figure 4. Average cross-price elasticities across pedigree groups. Each cell reports the mean pairwise cross-price elasticity between stallions in pedigree group g and pedigree group h, averaged across stallion pairs and overlapping market-years under the preferred nested-logit specification. Pairwise elasticities are first averaged across overlapping years and then symmetrized. Darker diagonal elements indicate that substitution is more concentrated within lineage-based groups than across them.

The elasticity calculations use two distinct conventions. Market-share effects are evaluated for representative stallion-years at the 25th, 50th, and 75th percentiles of baseline market share. Own-price elasticities are computed for each stallion-year observation and then averaged across the sample. This distinction matters because market-share effects are counterfactual magnitudes at representative market positions, whereas elasticities summarize the distribution of price sensitivity across observed stallion-years.

E Forecasting Robustness

This appendix provides additional details on the forecasting exercise used to interpret the demand response to recent graded-stakes victories. The main text reports the forecasting estimates for the baseline maturity horizon and two maturity-window checks. To avoid repeating the main table, this appendix clarifies the construction of the forecasting sample, the role of the exposure offset, and the interpretation of alternative future-output definitions.

E.1 Alternative maturity horizons

The forecasting exercise links recent success signals observed before year-t mating choices to the future elite output of the same year-t mating cohort. For each stallion-year (j, t), the outcome Y_{jt}^{H} counts the number of offspring from stallion j's year-t mating cohort whose first G1, G2, or G3 victory occurs by year $t + H$. Each offspring is counted at most once, at the time of its first graded-stakes victory. The exposure variable is the number of year-t matings, q_{jt}.

The baseline forecasting model is a Poisson count model with mating exposure as an offset:

$$Y_{jt}^{H} \sim \text{Poisson}(\mu_{jt}^{H}), \tag{43}$$

with

$$\log \mu_{jt}^{H} = \log q_{jt} + \lambda_1 G1_{jt} + \lambda_2 G2_{jt} + \lambda_3 G3_{jt} + \phi quality_{jt} + X_{jt}'\kappa + \tau_t. \tag{44}$$

The offset $\log q_{jt}$ adjusts for the number of mating opportunities. The coefficients therefore capture associations with future elite-output rates rather than raw future elite-output counts.

The maturity horizon H involves a tradeoff. A short horizon may miss late-developing offspring, while a long horizon reduces the usable sample because more recent mating cohorts have not yet had enough time to reveal elite outcomes. The main text uses $H = 5$ as the baseline and reports $H = 6$ and $H = 7$ as checks. Table 17 summarizes how the usable forecasting sample changes with the maturity horizon.

The main-text estimates show the same qualitative pattern across the three horizons. The latent-quality measure is positive and statistically significant for $H = 5$, $H = 6$,

Table 17. Forecasting sample by maturity horizon

	$H = 5$	$H = 6$	$H = 7$
Observations	3,594	3,375	3,170
Stallions	706	677	643
Future elite events	742	889	885
Mating exposure	179,384	167,771	156,364
Mating-year range	2004–2020	2004–2019	2004–2018
Mating-year fixed effects	Yes	Yes	Yes

Notes: The table summarizes the forecasting samples used for the maturity horizons reported in the main text. The dependent variable is the number of offspring from stallion j's year-t mating cohort whose first G1, G2, or G3 victory occurs by year $t + H$. Mating exposure is the sum of year-t matings in the corresponding forecasting sample.

and $H = 7$. Recent G1 and G2 victories are not statistically significant at any of these horizons. Recent G3 victories are positive but not statistically significant across the three horizons. Thus, the interpretation does not depend on a single maturity-window choice: durable quality measures forecast future elite output, while recent graded-stakes victories have limited robust incremental forecasting content once latent quality and other controls are included.

E.2 Alternative future-output definitions

The baseline outcome counts first G1, G2, or G3 victories from the year-t mating cohort within horizon H. This definition is designed to match the rare-success object used in the latent-quality measure. Counting only first graded-stakes victories avoids repeatedly counting the same offspring as multiple successes when the same horse wins several graded races over its career.

Alternative future-output definitions can be used to assess whether the limited incremental forecasting content of recent success signals depends on this exact outcome construction. Natural alternatives include: counting only future G1 victories; counting future G1 or G2 victories; counting all graded-stakes victories rather than first victories; varying the maturity window; or using a binary indicator for whether the year-t mating cohort produces at least one future graded-stakes winner.

These alternatives change the interpretation of the forecasting outcome. A G1-only outcome focuses on the most commercially visible elite successes but is sparser. A G1-or-G2 outcome remains closer to top-tier performance while retaining more observations.

Counting all graded-stakes victories allows repeated success by the same offspring to enter the outcome, but it no longer measures the arrival of first elite offspring. A binary cohort-success outcome emphasizes whether the mating cohort produces any elite winner, but discards information about the number of successful offspring.

The preferred outcome in the main text is therefore the first G1, G2, or G3 victory count. It is aligned with the empirical-Bayes quality measure, adjusts naturally for mating exposure, and preserves the interpretation of rare-success productivity as the rate at which matings generate first elite outcomes. These alternative definitions provide useful directions for additional sensitivity checks, but the current analysis focuses on the first G1, G2, or G3 victory count because it is aligned with the empirical-Bayes quality measure and preserves the interpretation of rare-success productivity as a first-arrival rate. The key robustness question is whether recent graded-stakes victories acquire stable incremental forecasting content once the future-output definition is changed. The current evidence supports the main interpretation: latent quality is the more robust predictor of future elite output, while recent success signals are more important for demand than for incremental forecasting.

F Output-Alignment and Diversion Robustness

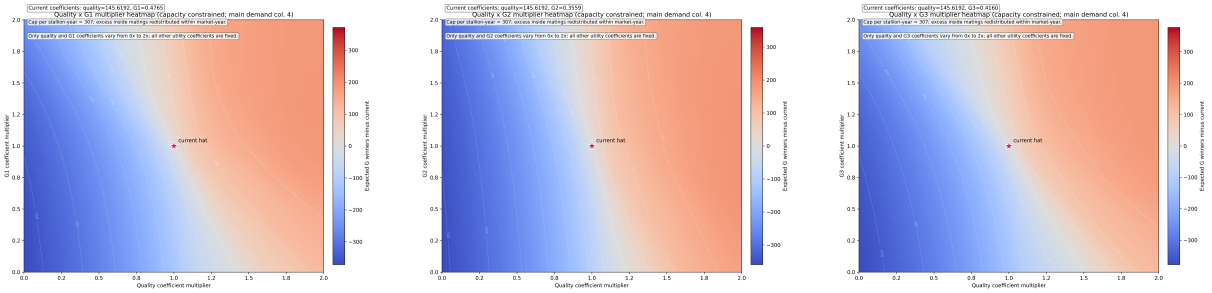
This appendix provides additional details and robustness checks for the output-alignment and signal-induced diversion exercises. The main text emphasizes the representative-breeder output-alignment diagnostic and reports the G1-induced diversion decomposition. This appendix reports supplementary by-grade output-alignment figures and additional signal-induced focal-gain calculations.

F.1 Output-alignment by grade

The main text varies a common demand weight on recent G1, G2, and G3 success signals because that exercise corresponds most closely to the paper's central distinction between latent quality and salient recent success. This subsection reports a by-grade version of the same idea. Each panel varies the demand weight on latent quality and one recent graded-stakes success signal, holding the remaining recent-signal coefficients and all other utility components fixed.

The figures in this subsection should be read as supplementary output-alignment evi-

dence, not as a replacement for the representative-breeder exercise in the main text.



(a) $G1_{jt}$ (b) $G2_{jt}$ (c) $G3_{jt}$

Figure 5. Output alignment by grade of recent success signal. Each panel varies the demand weight on latent quality on the horizontal axis and the demand weight on one recent graded-stakes success signal on the vertical axis. The color scale reports the change in expected future elite output relative to the allocation implied by the estimated demand weights in the supplementary capacity-capped allocation diagnostic. Prices, product sets, pedigree nests, market-year fixed effects, and all other utility components are held fixed.

Figure 5 shows that the qualitative pattern is not driven by a single grade of recent success. Across the G1, G2, and G3 panels, output alignment improves when demand weight shifts toward latent quality relative to the individual recent success signal. This pattern supports the main-text interpretation that latent quality is more closely aligned with future rare-success outcomes than salient recent realizations.

F.2 Diversion of G2- and G3-induced demand gains

The main text reports the diversion-style decomposition for G1-induced demand gains because G1 is the most visible and economically important recent success signal. The same computation can be applied to G2 and G3 signals. For a focal stallion-year with a recent grade-k signal, where $k \in \{2, 3\}$, the focal signal-induced demand gain is

$$\Delta s_{jt}^{k} = \widehat{s}_{jt} - \widehat{s}_{jt}^{(-k)}, \tag{45}$$

where $\widehat{s}_{jt}$ is the predicted share under the estimated demand system and $\widehat{s}_{jt}^{(-k)}$ is the predicted share after removing the direct utility contribution of the grade-k recent signal for the focal stallion.

Table 18 shows that lower-grade recent success signals also generate economically meaningful demand gains. The focal gain is smaller for G2 and G3 than for G1, but the effects

Table 18. Signal-induced focal demand gains by grade

Recent signal	Focal gain (pp)
$G1_{jt}$	0.565
$G2_{jt}$	0.449
$G3_{jt}$	0.363

Notes: The table reports the average increase in the focal stallion's predicted market share from the direct recent-signal utility component, measured in percentage points. The G1 source decomposition is reported in the main text. The G2 and G3 calculations apply the same counterfactual logic to lower-grade recent success signals.

are still large relative to typical stallion-year market shares. These calculations support the interpretation that recent graded-stakes victories act as salient demand shifters across the grade hierarchy.

G Pricing Robustness

This appendix reports additional pricing specifications. The main text presents the baseline within-stallion posted-fee capitalization regression. The appendix collects robustness checks that support the reduced-form pricing interpretation without overloading the main text.

G.1 Annual fee revisions

The main text studies within-stallion movements in posted-fee levels. A complementary specification focuses directly on year-to-year fee revisions. The first-difference specification is

$$\Delta \log(1+p_{jt}) = \lambda_1 G1_{jt} + \lambda_2 G2_{jt} + \lambda_3 G3_{jt} + \psi \Delta quality_{jt} + \chi \Delta serviceyear_{jt} + \tau_t + u_{jt}. \quad (46)$$

Recent graded-stakes victories enter in levels because they represent new public information available before the fee revision. Latent quality and service year enter in first differences because the specification focuses on annual changes in the persistent state variables.

Table 19 supports the main reduced-form interpretation. Recent G1 victories remain positively associated with annual fee revisions, and changes in latent quality are also positively associated with fee revisions. The G2 and G3 coefficients are less stable in this specification. For this reason, the main text emphasizes selective capitalization, with the most robust pricing response concentrated in G1 victories and latent quality.

Table 19. First-difference pricing response

	$\Delta \log(1 + p_{jt})$
$G1_{jt}$	0.0627***
	(0.0143)
$G2_{jt}$	-0.0075
	(0.0122)
$G3_{jt}$	-0.0149
	(0.0104)
$\Delta quality_{jt}$	18.2135***
	(6.2569)
Δ Service year	-0.3994*
	(0.2302)
Observations	3,546
Year fixed effects	Yes
Clustered standard errors	Stallion

Notes: The dependent variable is the year-to-year change in $\log(1 + p_{jt})$. Recent G1, G2, and G3 victories enter in levels, while latent quality and service year enter in first differences. Stallion fixed effects are removed by differencing, and year fixed effects are included in the differenced regression. Each observation is matched to its prior calendar year, so the dependent variable represents an annual fee revision only when the immediately preceding year is observed. Standard errors clustered at the stallion level are reported in parentheses. Significance levels are denoted by $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

G.2 Timing checks

The baseline pricing interpretation relies on the market timing: progeny outcomes are observed before the following breeding season, after which posted fees are announced. This subsection reports auxiliary lead-lag checks that add exact within-stallion lead and lag controls for recent graded-stakes victories.

Table 20 should be read as a timing diagnostic rather than as a separate causal design. The current and lagged G1 terms are positive and statistically significant, consistent with posted fees adjusting around publicly observed top-grade progeny success. The G2 and G3 timing patterns are weaker. This reinforces the main-text interpretation that pricing responses are most robust for the most visible recent success signal.

G.3 Alternative coding of recent success signals

The baseline pricing specification uses counts of recent G1, G2, and G3 victories. This subsection reports two alternative codings. The first replaces counts with indicators for whether at least one recent victory of each grade occurred. The second uses mutually

Table 20. Timing checks for pricing responses to graded-stakes victories

	G1 lead/lag	All graded lead/lag
$G1_{j,t+1}$	0.0494**	0.0428*
	(0.0251)	(0.0232)
$G1_{jt}$	0.1133***	0.1099***
	(0.0274)	(0.0264)
$G1_{j,t-1}$	0.1296***	0.1247***
	(0.0344)	(0.0331)
$G2_{j,t+1}$		-0.0102
		(0.0272)
$G2_{jt}$		0.0272
		(0.0284)
$G2_{j,t-1}$		-0.0125
		(0.0357)
$G3_{j,t+1}$		0.0166
		(0.0183)
$G3_{jt}$		0.0194
		(0.0223)
$G3_{j,t-1}$		-0.0175
		(0.0248)
$quality_{jt}$	30.1668*	29.6304
	(16.7993)	(19.5574)
Service year	-0.1336	-0.1273
	(0.4845)	(0.4787)
Observations	2,825	2,825
Stallion fixed effects	Yes	Yes
Market-year fixed effects	Yes	Yes
Clustered standard errors	Stallion	Stallion

Notes: The dependent variable is $\log(1+p_{jt})$. Column (1) adds exact one-year lead and lag terms for G1 victories within stallion; column (2) adds the corresponding G2 and G3 lead-lag terms. Lead and lag variables are constructed by matching observations within stallion across adjacent market years, so the exact-match lead-lag sample is smaller than the baseline fixed-effects sample. All columns absorb stallion and market-year fixed effects. Standard errors clustered at the stallion level are reported in parentheses. Significance levels are denoted by $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

exclusive indicators for the best recent signal: G1, G2 only, or G3 only, with no recent graded victory as the omitted category.

Table 21. Robustness to alternative coding of recent success signals

	Baseline	Any-win	Best-signal
$G1_{jt}$	0.1270***		
	(0.0365)		
$G2_{jt}$	0.0661**		
	(0.0281)		
$G3_{jt}$	0.0417		
	(0.0265)		
$\mathbf{1}\{G1_{jt} > 0\}$		0.3083***	
		(0.0566)	
$\mathbf{1}\{G2_{jt} > 0\}$		0.1179**	
		(0.0486)	
$\mathbf{1}\{G3_{jt} > 0\}$		0.1381***	
		(0.0381)	
Best recent signal = G1			0.4030***
			(0.0604)
Best recent signal = G2 only			0.1317**
			(0.0561)
Best recent signal = G3 only			0.1164***
			(0.0424)
$quality_{jt}$	39.6984**	47.4748***	53.3702***
	(16.8288)	(14.2629)	(14.1228)
Service year	-0.0222	-0.0163	-0.0189
	(0.1852)	(0.1839)	(0.1857)
Observations	4,467	4,467	4,467
Stallion fixed effects	Yes	Yes	Yes
Market-year fixed effects	Yes	Yes	Yes
Clustered standard errors	Stallion	Stallion	Stallion

Notes: The dependent variable is $\log(1+p_{jt})$. Column (1) reproduces the baseline count specification for comparison. Column (2) replaces graded-win counts with indicators for whether at least one G1, G2, or G3 victory occurred. Column (3) replaces the counts with mutually exclusive indicators for the best recent signal: G1, G2 only, or G3 only, leaving no recent graded victory as the omitted category. All columns use the same fixed-effects sample, absorb stallion and market-year fixed effects, and report standard errors clustered at the stallion level in parentheses. Significance levels are denoted by $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 21 shows that the qualitative pricing pattern is not driven by using count variables. Indicator and best-signal codings also show positive capitalization of recent success signals, with the strongest response concentrated in G1. Latent quality remains positively associated with posted fees across these specifications.

G.4 Alternative fee transformations

The baseline outcome is $\log(1+p_{jt})$, which allows zero posted fees to remain in the sample. This subsection reports robustness checks using alternative fee transformations: $\log(p_{jt})$ for strictly positive prices and $\text{asinh}(p_{jt})$ for the full sample.

Table 22. Robustness to alternative fee transformations

	Baseline $\log(1+p)$	Positive-price $\log p$	$\text{asinh}(p)$
$G1_{jt}$	0.1270***	0.1277***	0.1274***
	(0.0365)	(0.0367)	(0.0368)
$G2_{jt}$	0.0661**	0.0660**	0.0658**
	(0.0281)	(0.0283)	(0.0285)
$G3_{jt}$	0.0417	0.0423	0.0422
	(0.0265)	(0.0267)	(0.0268)
$quality_{jt}$	39.6984**	43.0999**	40.9812**
	(16.8288)	(17.0150)	(17.0811)
Service year	-0.0222	-0.0069	-0.0150
	(0.1852)	(0.2065)	(0.1905)
Observations	4,467	4,354	4,467
Stallion fixed effects	Yes	Yes	Yes
Market-year fixed effects	Yes	Yes	Yes
Clustered standard errors	Stallion	Stallion	Stallion

Notes: Each column keeps the baseline regressor set but varies the outcome transformation. Column (1) uses $\log(1+p_{jt})$. Column (2) restricts the sample to strictly positive stud fees and uses $\log(p_{jt})$; relative to the baseline fixed-effects sample, this drops 113 zero-fee observations. Column (3) uses $\text{asinh}(p_{jt})$ on the full sample. All columns absorb stallion and market-year fixed effects, and standard errors clustered at the stallion level are reported in parentheses. Significance levels are denoted by $^{*}p<0.10$, $^{**}p<0.05$, and $^{***}p<0.01$.

Table 22 shows that the baseline pricing result is not driven by the specific $\log(1+p_{jt})$ transformation. The G1 and G2 coefficients remain positive and statistically significant under the positive-price log specification and the inverse hyperbolic sine transformation. The G3 coefficient remains smaller and less stable. Latent quality remains positively associated with posted fees across the alternative transformations.